This demo contains:

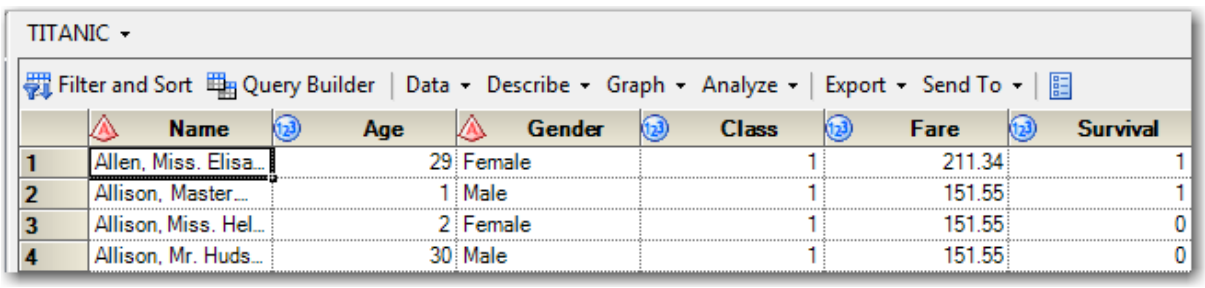
- **Examining Distributions**
- **Creating Two-Way Frequency Tables**
- **Chi-Square Test**
- **Detecting Ordinal Associations**
- **Simple Logistic Regression Model**
- **Multiple Logistic Regression with Categorical Predictors**
- **Logistic Regression Backward Elimination with Interactions**

Examining Distributions

Example: Use the One-Way Frequencies task to create one-way frequency tables for the variables **Survival**, **Gender**, and **Class** in the **Titanic** data set.

1. Create a new process flow and rename it **Categorical Data Analysis and Logistic Regression**.

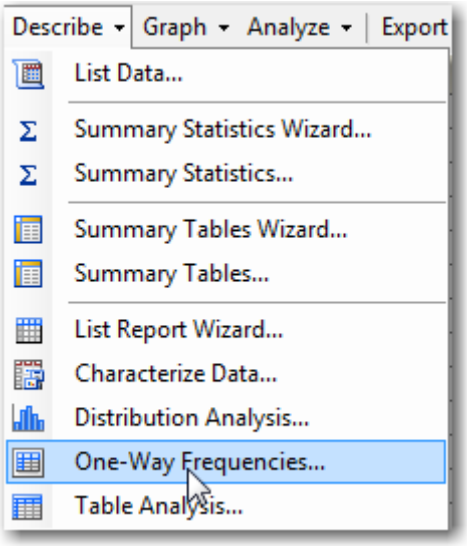
2. Open the **Titanic** data set from the SASUSER library on the Local server.



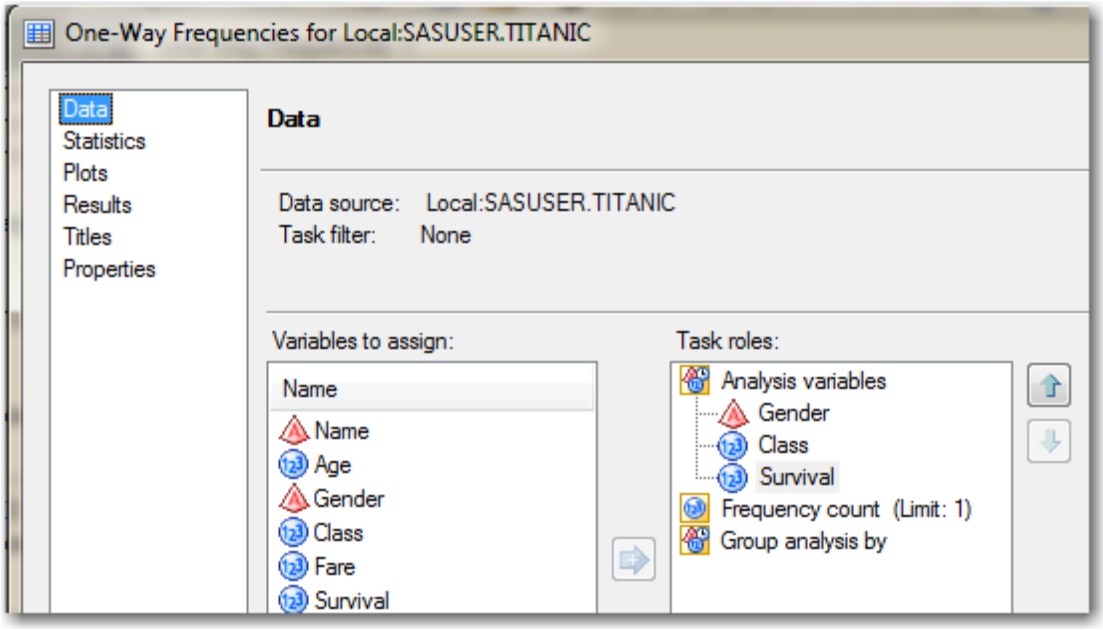
	Name	Age	Gender	Class	Fare	Survival
1	Allen, Miss. Elisa...	29	Female	1	211.34	1
2	Allison, Master...	1	Male	1	151.55	1
3	Allison, Miss. Hel...	2	Female	1	151.55	0
4	Allison, Mr. Huds...	30	Male	1	151.55	0

Create the frequency tables for the discrete variables.

3. Select **Describe** ⇨ **One-Way Frequencies** from the menu above the data table.



4. Add **Gender**, **Class**, and **Survival** to the analysis variables role.



5. Click **Run**.

Output

Gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	466	35.60	466	35.60
Male	843	64.40	1309	100.00

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

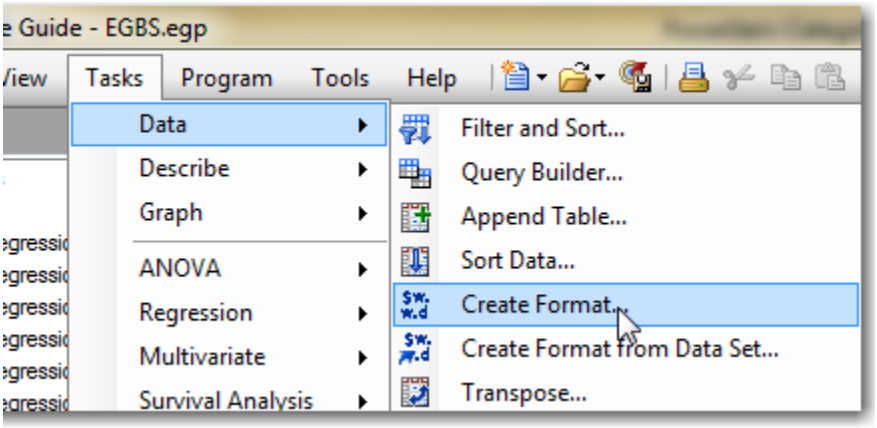
Class	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	323	24.68	323	24.68
2	277	21.16	600	45.84
3	709	54.16	1309	100.00

Survival	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	809	61.80	809	61.80
1	500	38.20	1309	100.00

There seem to be no unusual data values that could be due to coding errors for any of the categorical variables.

The **Survival** variable is internally coded **`0`** for those who died and **`1`** for those who survived the Titanic crash. Create a format that assigns a value **`Survived`** to observations whose internal value is **`1`** and **`Died`** to observations whose internal value is **`0`**.

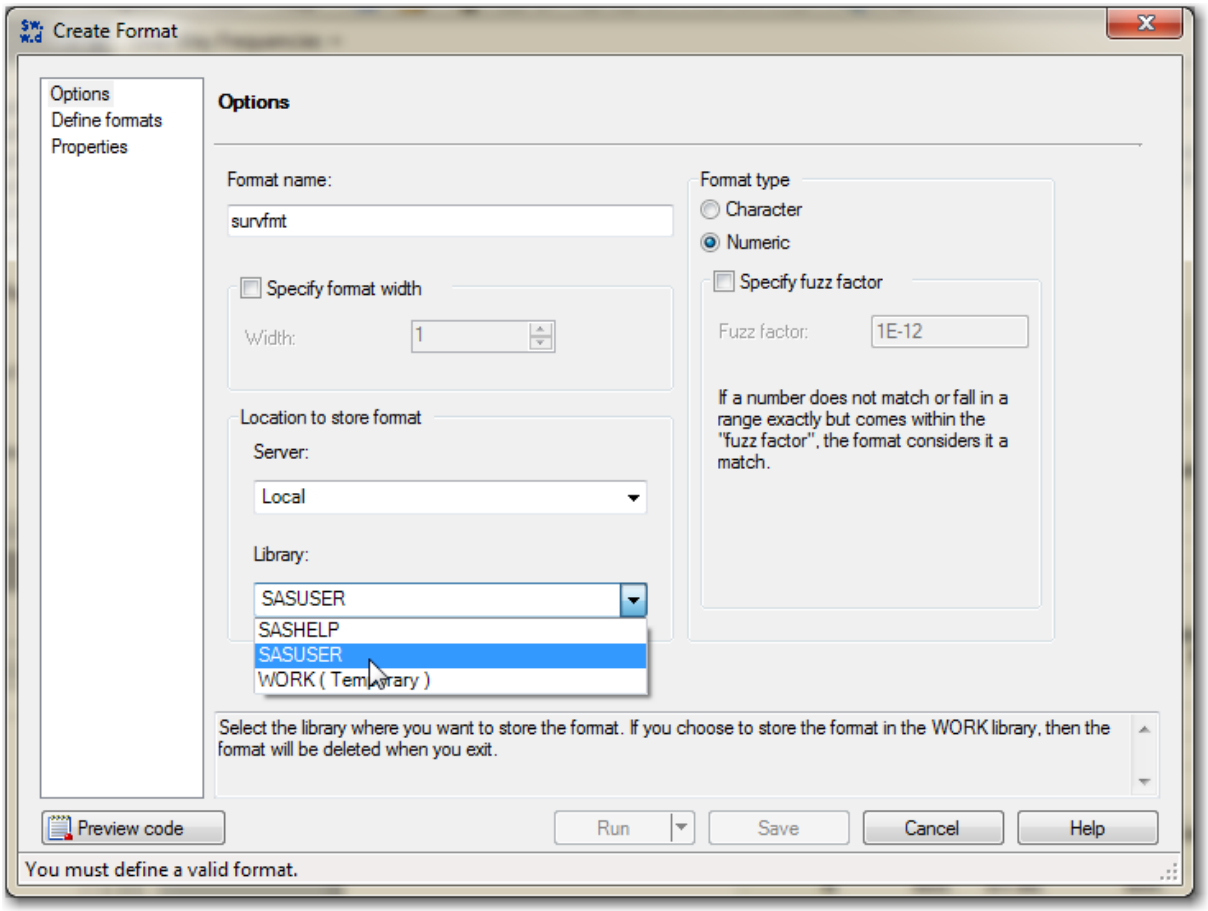
6. Select **Tasks** ⇨ **Data** ⇨ **Create Format**.



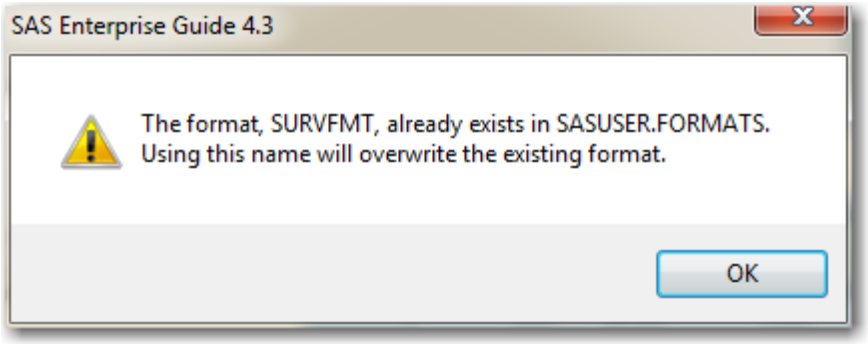
Formats are created independently of the data. You do not need to select a data set before you create a format.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

7. Type **survfmt** as the format name. Select **Numeric** because this format will be applied to a numeric variable. Select **SASUSER** as the library where you would like to save your format.

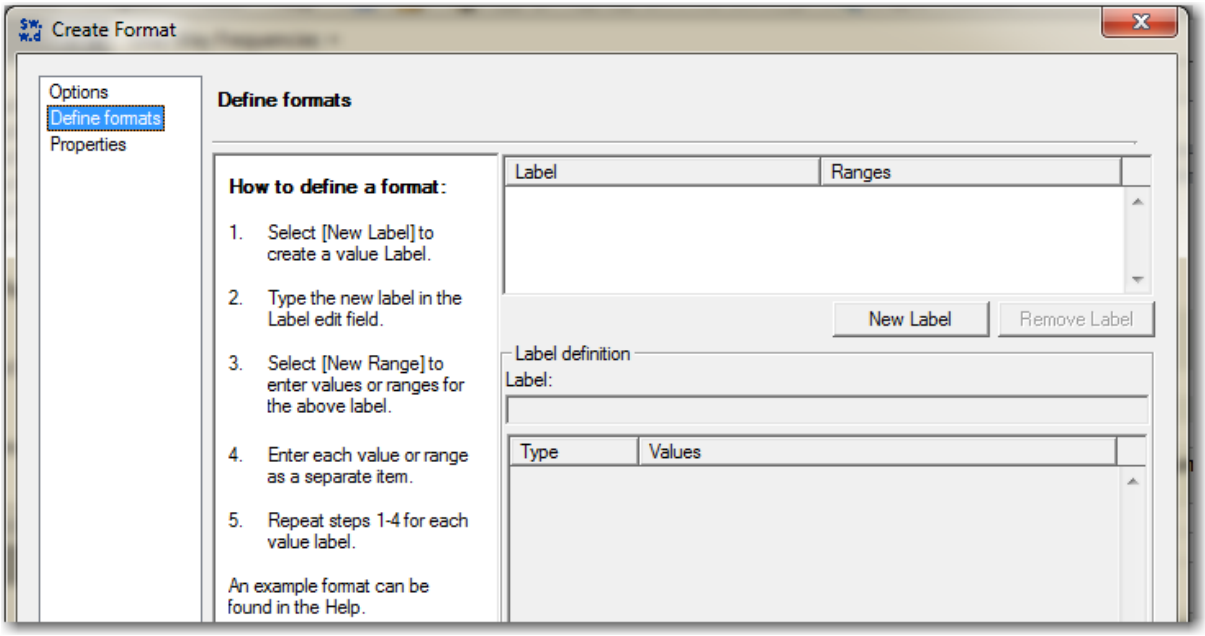


A dialog box might open because a format of the same name was created in the data creation file submitted at the beginning of the course. Click **OK** and continue.

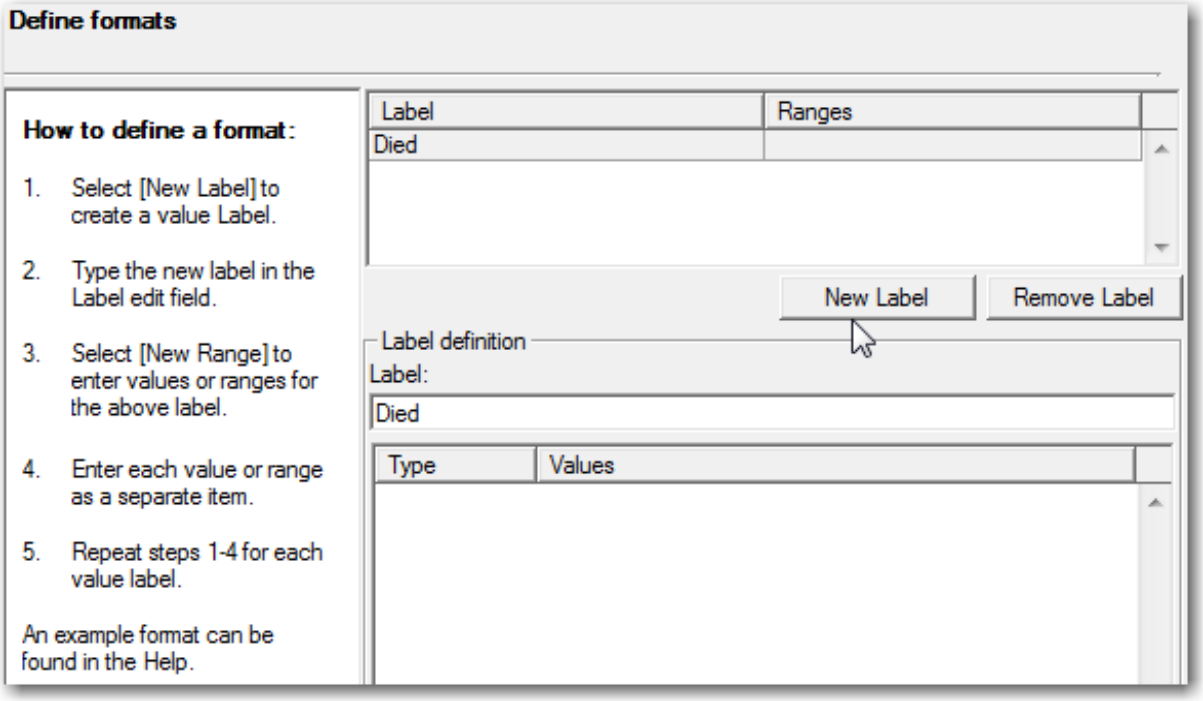


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

8. Select **Define formats** at the left and follow the instructions on how to define a format.

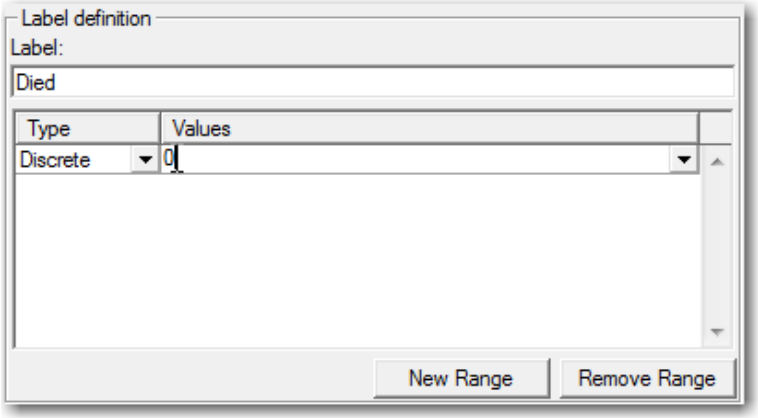


9. Click **New Label** and then type **Died** as the label.



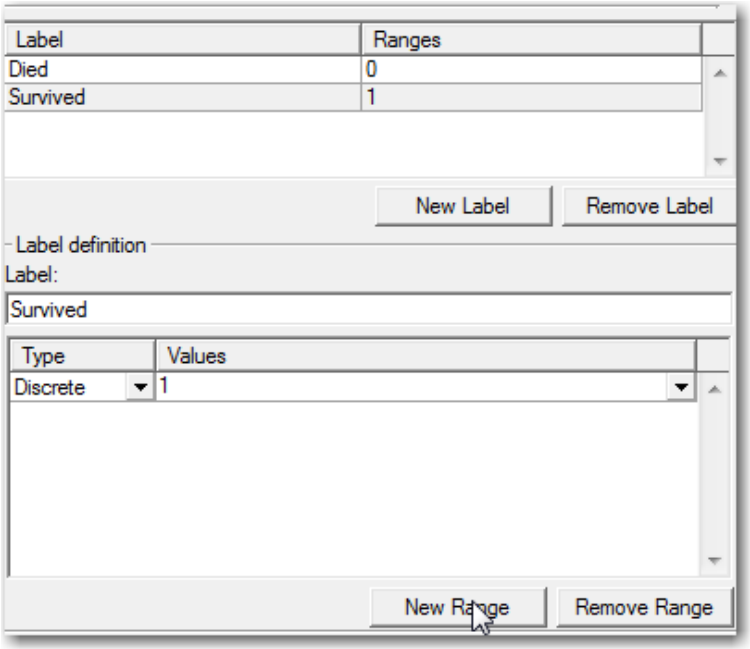
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

10. Click **New Range** to add format values to assign to internal levels of a variable. Keep the type as **Discrete** and then type **0** as the value.

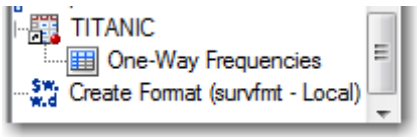


11. Click **New Label** again and then type **Survived** as the label.

12. Click **New Range**. Keep the type as **Discrete** and then type **1** as the value.



13. Click **Run**.



The format is now ready to use.

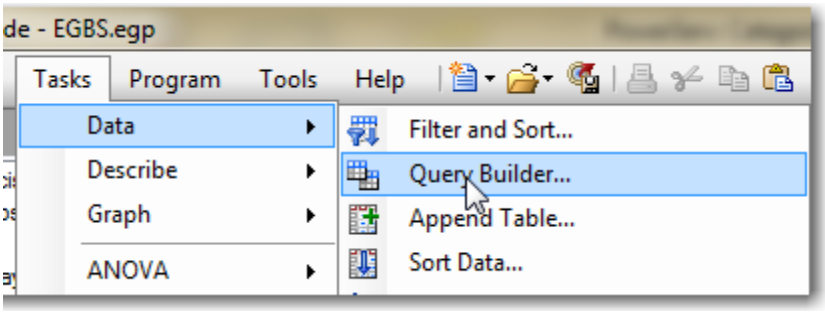
The format CLASSFMT. was also created in the SASUSER library in the data creation file at the beginning of the course.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

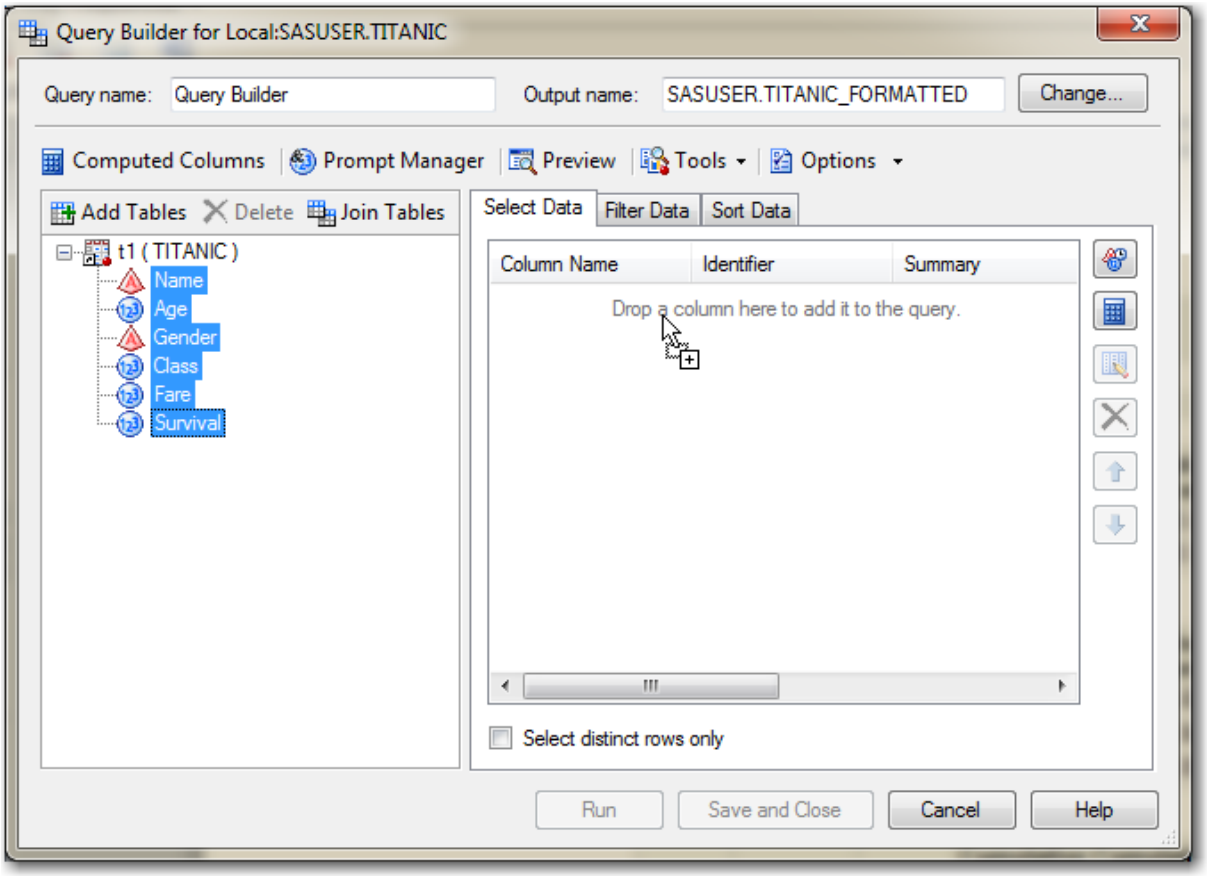
The formats CLASSFMT and SURVFMT have been created but they have not yet been associated with any variables. The associations can be made in each task run. They can also be made during the creation of a data table. Create the associations by creating a copy of the Titanic data set.

14. Verify that **Titanic** is the active data source by selecting it in the **Process Flow** or **Project Explorer**.

15. Select **Tasks** ⇨ **Data** ⇨ **Query Builder**.

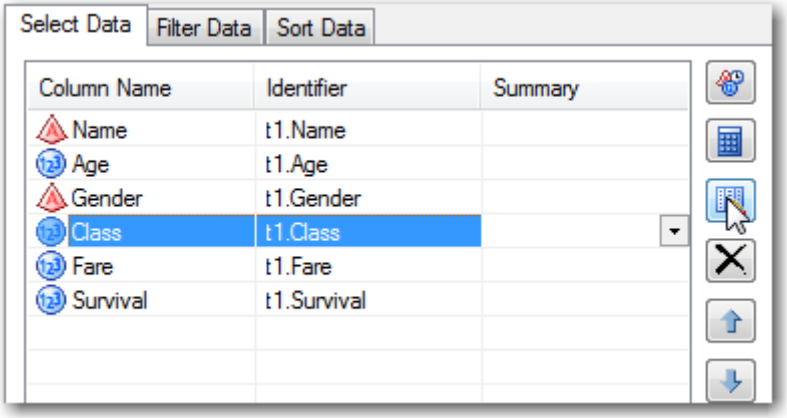


16. Drag all variables from the left pane to the right pane.

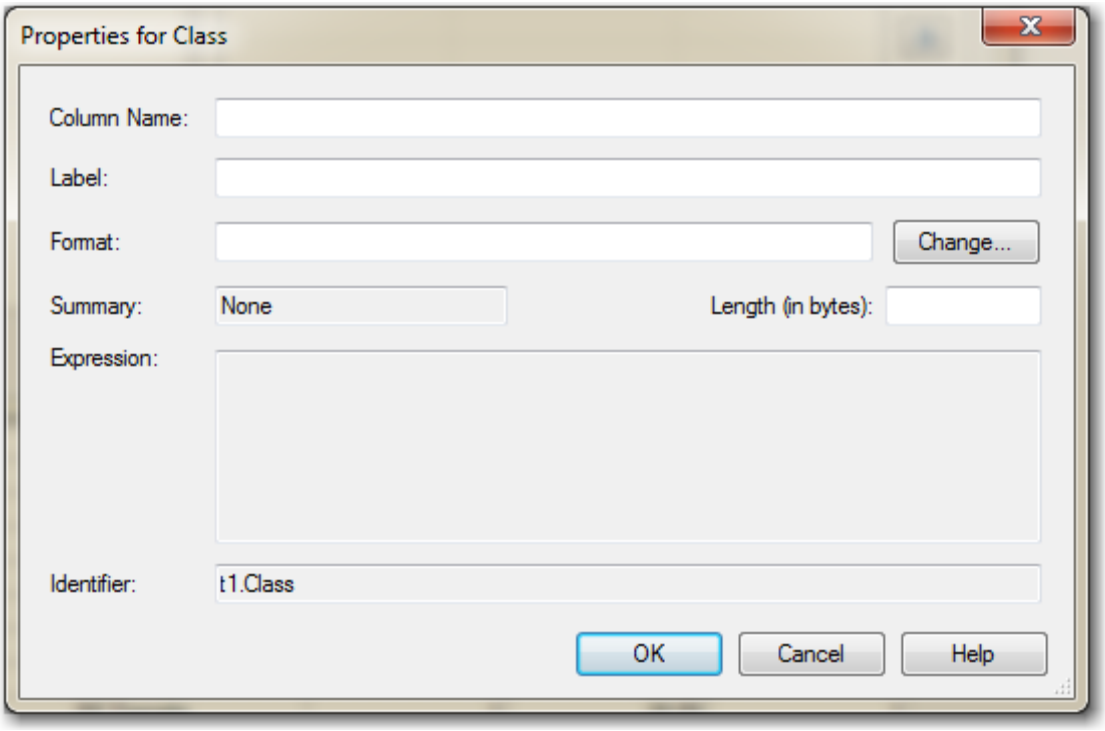


17. Select **Class** under the **Column Name** list, and then click at the right.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



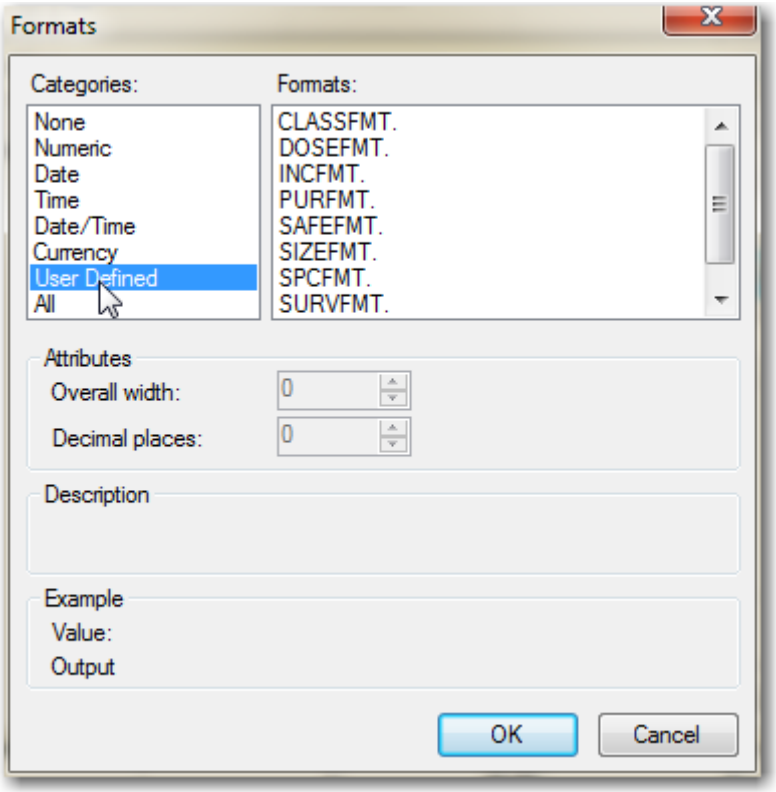
18. A Properties window appears for **Class**.



19. Click **Change** next to the text box for **Format**.

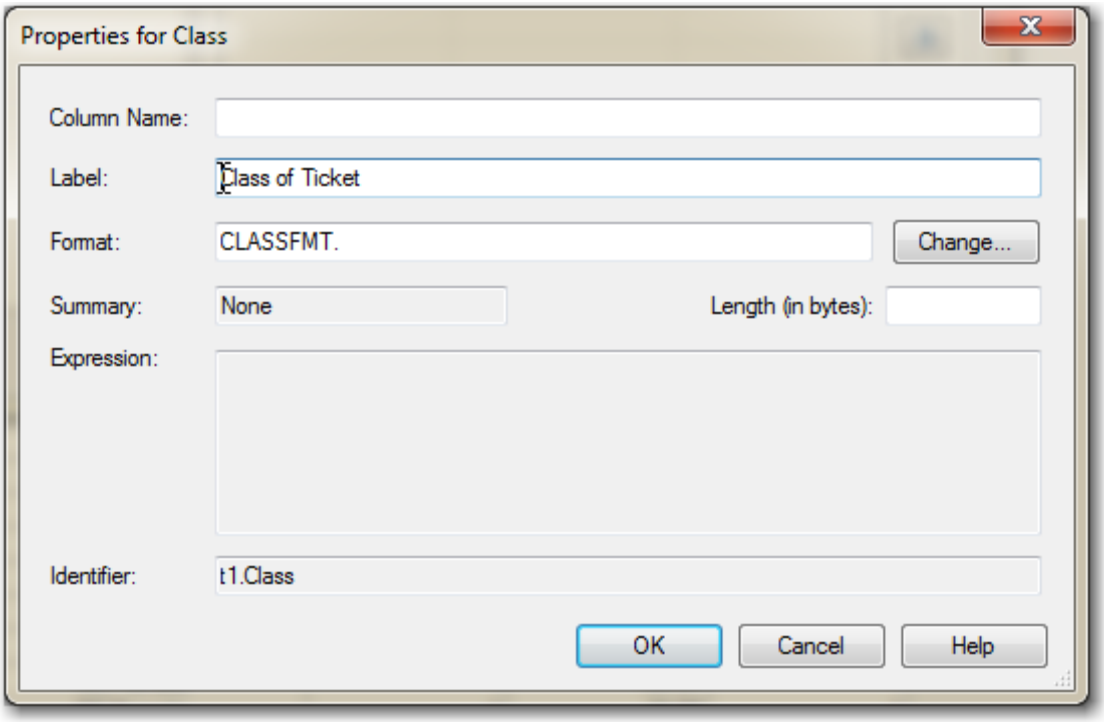
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

20. In the window that appears, select **User Defined** under **Categories**.



21. Highlight **CLASSFMT.** from the **Formats**: list and then click **OK**.

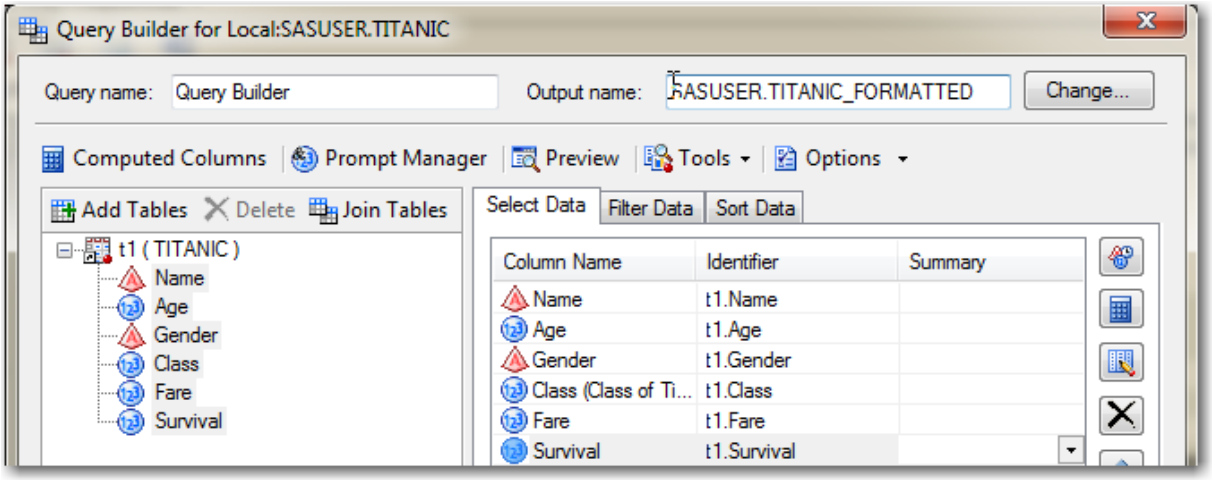
22. Change the label to **Class of Ticket**.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

23. Click **OK**.

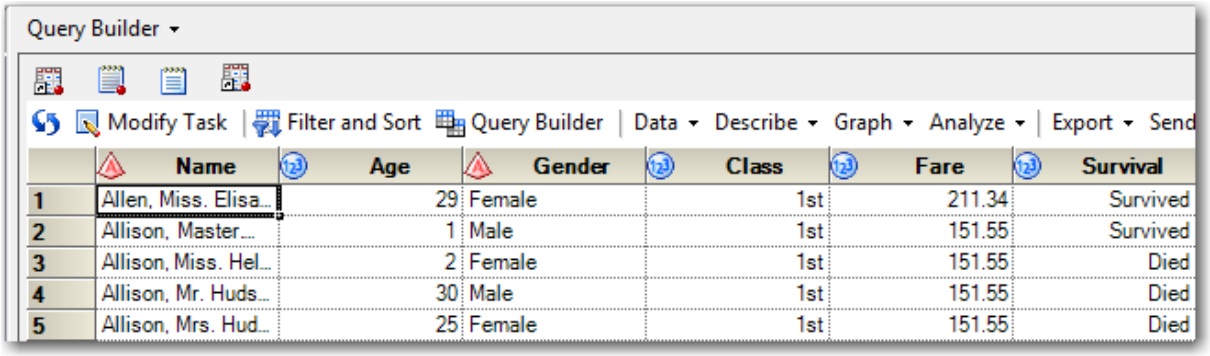
24. Change the output name in the upper right corner to **SASUSER.TITANIC_FORMATTED**.



25. Click the **Save** button.

26. Repeat steps 4 through 8 for the variable **Survival**. Apply the **SURVFMT**. format. Change the label if desired.

27. Click **Run**.



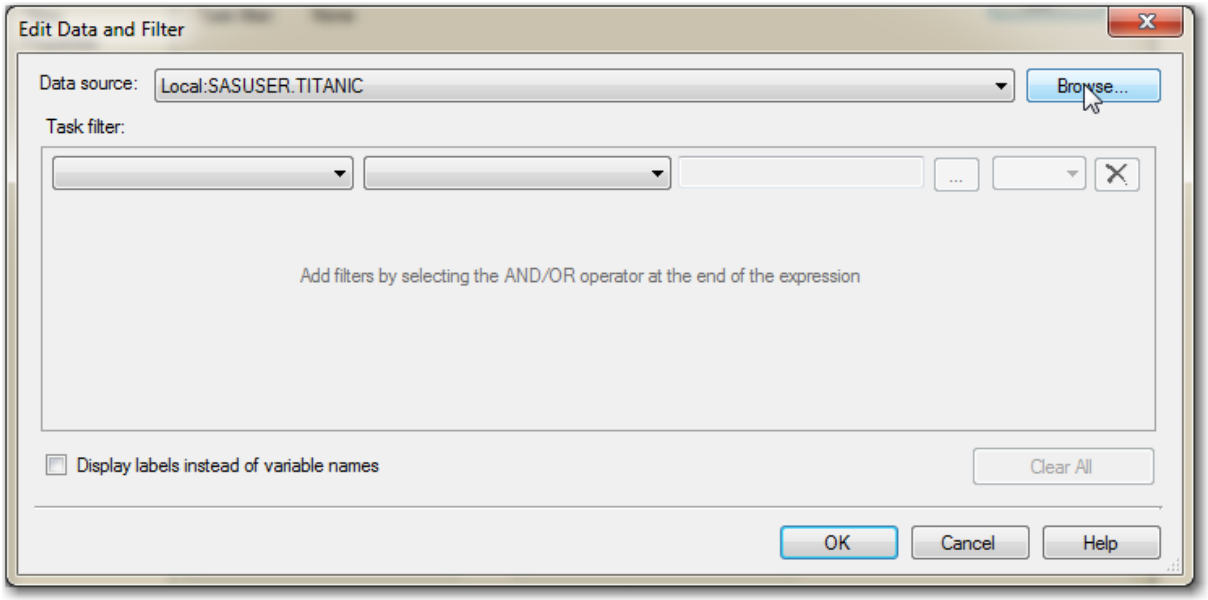
Rerun the previous analysis on the **TITANIC_FORMATTED** data set.

28. Modify the previous One-Way Frequencies task.

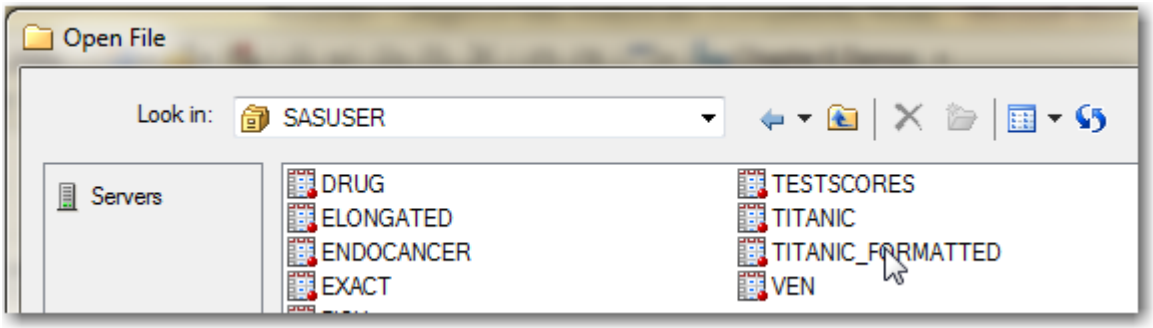
29. Click **Edit** at the upper right of the task window. This enables you to change the data set that the task runs on.

30. Click **Browse** to change the input data set.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



31. Navigate to the SASUSER library (**Local** ⇨ **Libraries** ⇨ **SASUSER**) and double-click the **Titanic_Formatted** data set.



32. Click **OK** to close the Edit Data and Filter window.

33. Click **Run**. Do not replace the results from the previous run.

Output

Gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	466	35.60	466	35.60
Male	843	64.40	1309	100.00

Class of Ticket				
Class	**Frequency**	**Percent**	**Cumulative Frequency**	**Cumulative Percent**
1st	323	24.68	323	24.68
2nd	277	21.16	600	45.84
3rd	709	54.16	1309	100.00

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Survival	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Died	809	61.80	809	61.80
Survived	500	38.20	1309	100.00

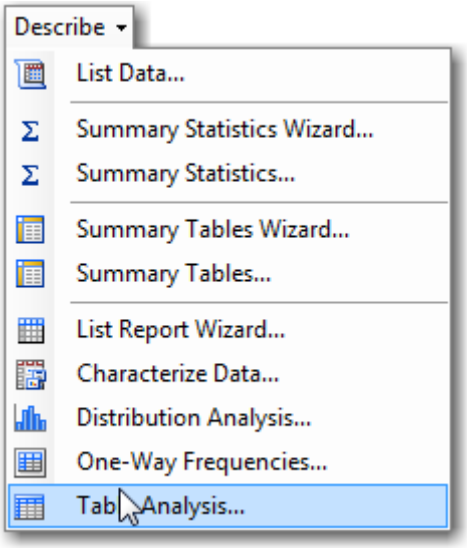
The table statistics have not changed.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

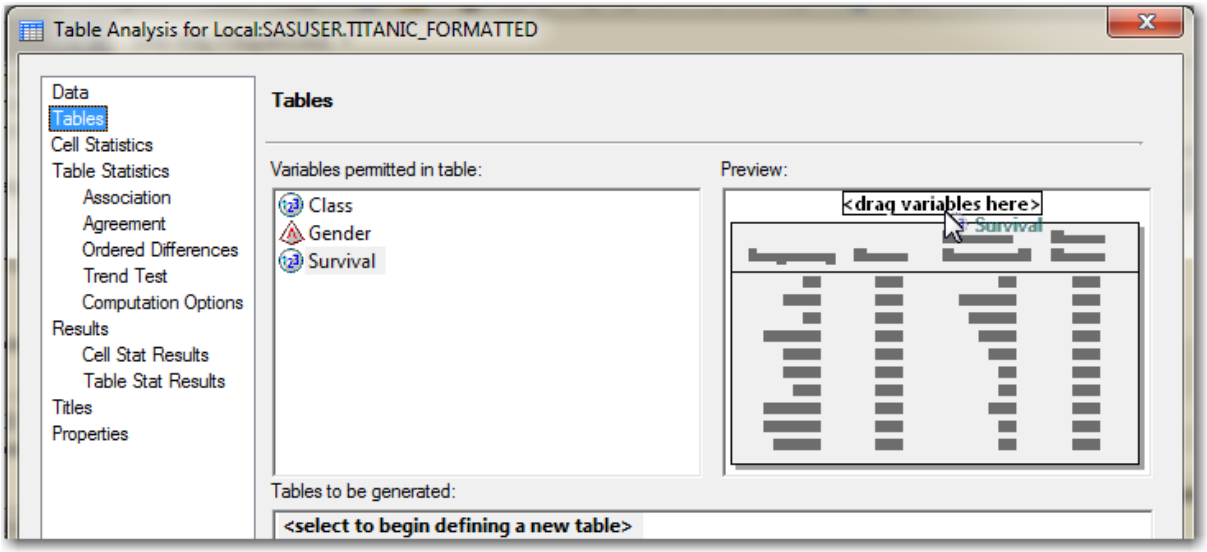
Creating Two-Way Frequency Tables

Create crosstabs (two-way frequency tables) to assess possible categorical associations with **Survival**.

1. Select the **Titanic_Formatted** data set.
2. Select **Describe** ⇨ **Table Analysis**.

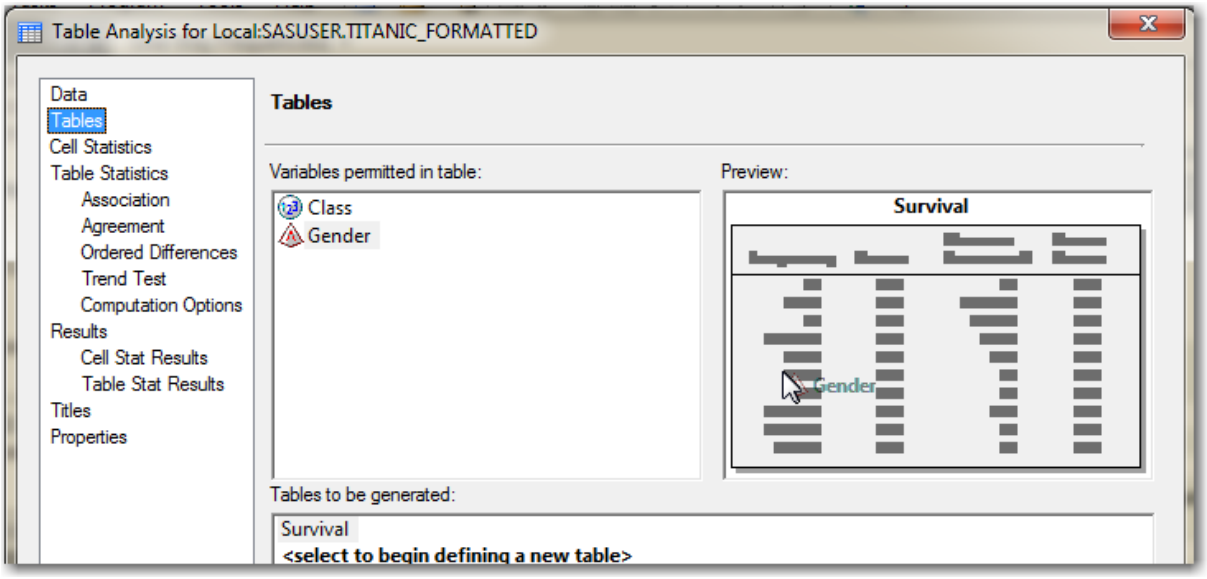


3. Assign **Gender**, **Class**, and **Survival** to the **Table variables** task role.
4. Select **Tables** at the left.
5. Drag **Survival** to the **Preview** area over the words **<drag variables here>**.

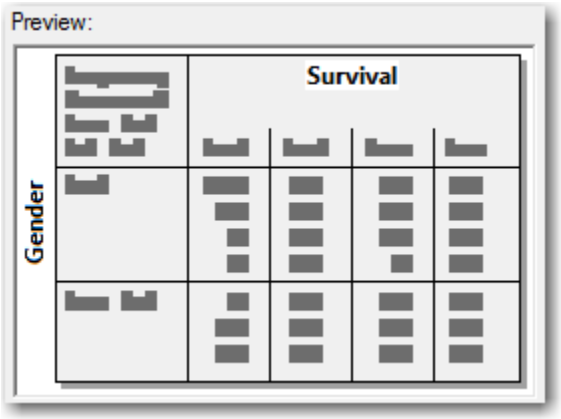


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. Drag **Gender** to the gray table template area.

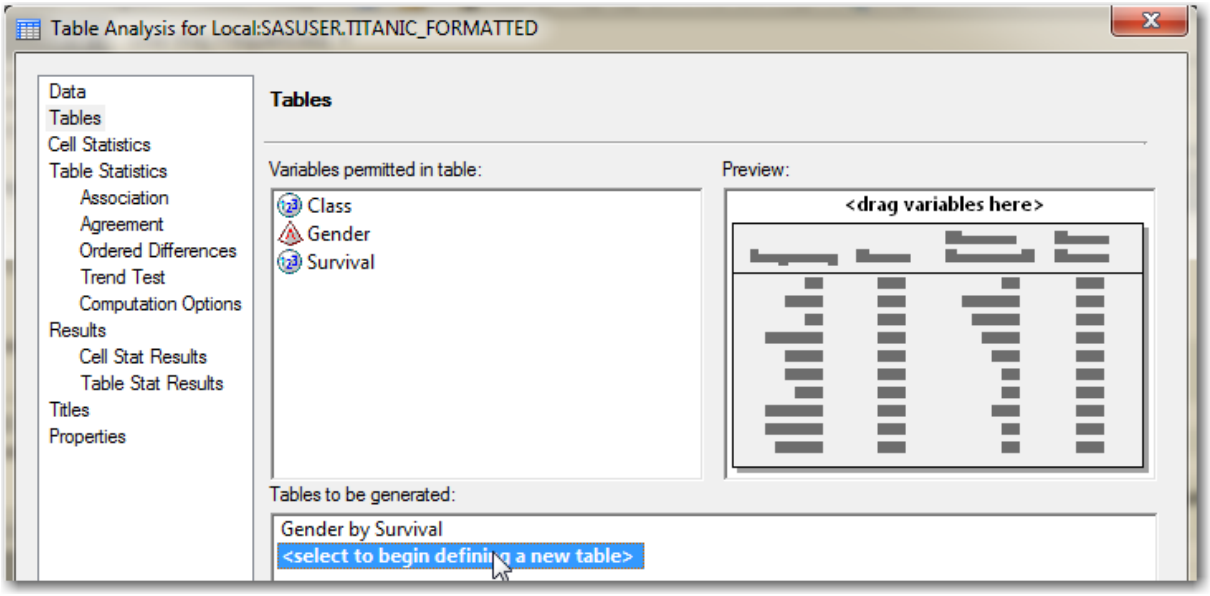


7. The table should show **Survival** at the top and **Gender** at the left.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

8. Click the words **<select to begin defining a new table>**.

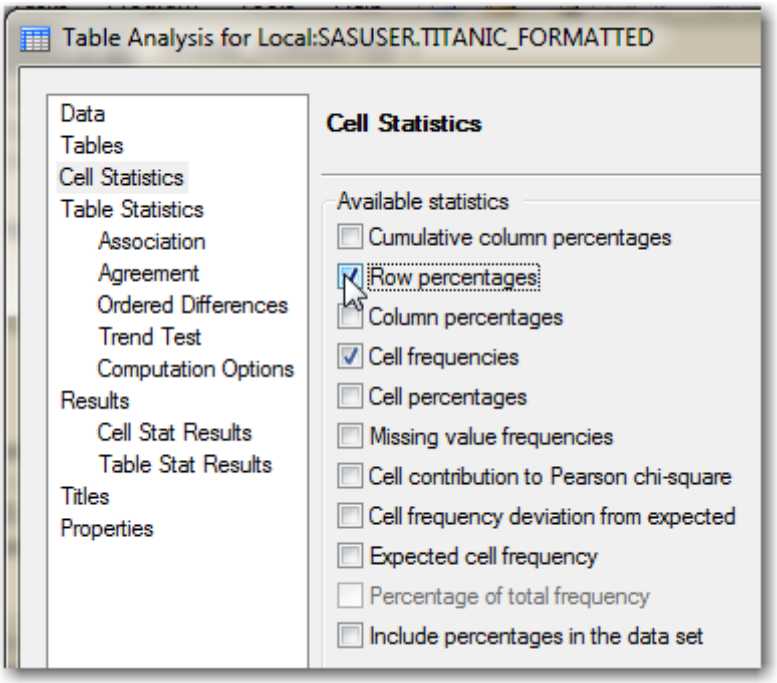


9. Drag **Survival** to the **Preview** area over the words **<drag variables here>**.

10. Drag **Class** to the gray table template area.

11. The table should show **Survival** at the top and **Class** at the left.

12. Select **Cell Statistics** at the left and then check the box for **Row percentages** and uncheck the box for **Column percentages**.



13. Click **Run**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The requested two-way frequency tables are shown below. You can get a preliminary idea whether there are associations between the outcome variable, **Survival**, and the predictor variables, **Gender** and **Class**, by examining the distribution of **Survival** at each value of the predictors.

Table of Gender by Survival			
Gender	**Survival**		
Frequency Row Pct	**Died**	**Survived**	**Total**
Female	127 27.25	339 72.75	466
Male	682 80.90	161 19.10	843
Total	809	500	1309

By examining the row percentages, you see that **Survived** is associated with **Gender**.

Table of Class by Survival			
Class(Class of Ticket)	**Survival**		
Frequency Row Pct	**Died**	**Survived**	**Total**
1st	123 38.08	200 61.92	323
2nd	158 57.04	119 42.96	277
3rd	528 74.47	181 25.53	709
Total	809	500	1309

There also seems to be an association between **Survival** and **Class**, with a far greater chance of surviving in higher classes.

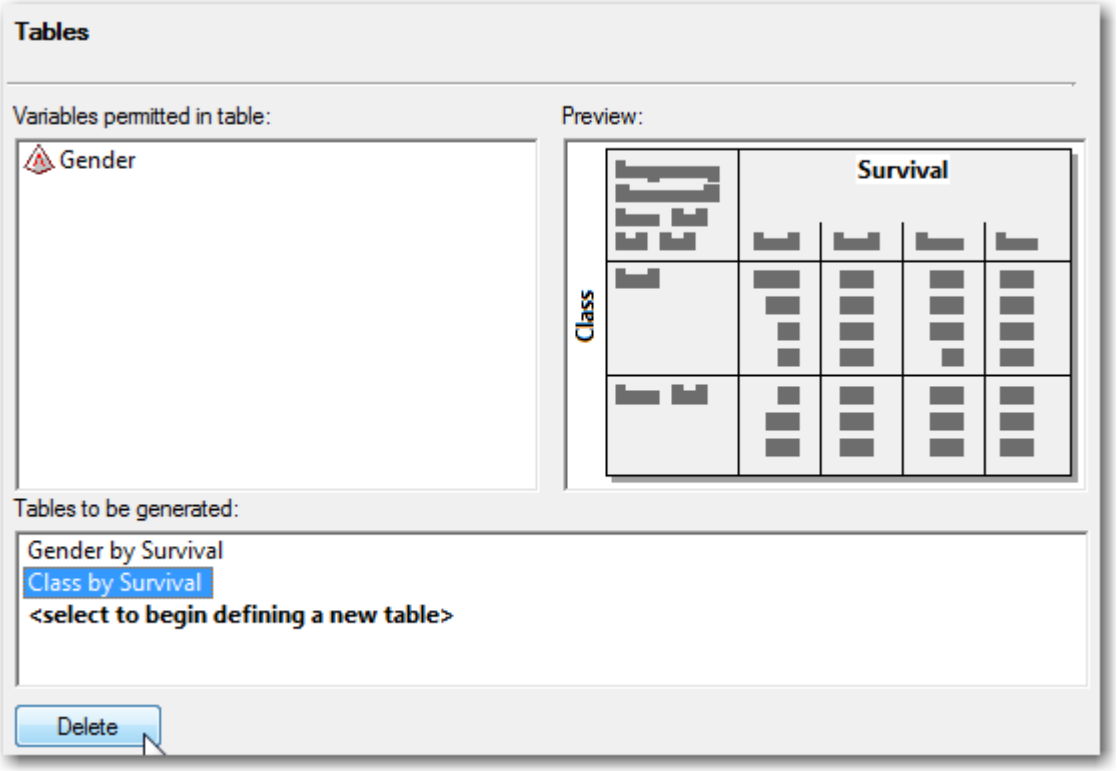
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Chi-Square Test

Example: Use the Table Analysis task to test for an association between the variables **Gender** and **Survived**. Also generate the expected cell frequencies and the cell's contribution to the total chi-square statistic.

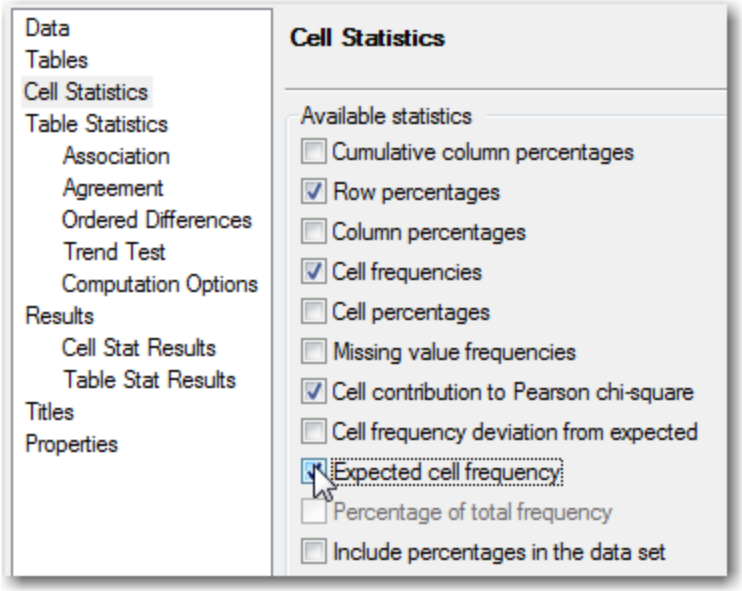
1. Reopen the Table Analysis task last created by right-clicking it and selecting **Modify** from the drop-down menu.

2. Select **Tables** at the left and delete the **Class** by **Survival** table by selecting it and clicking **Delete**.

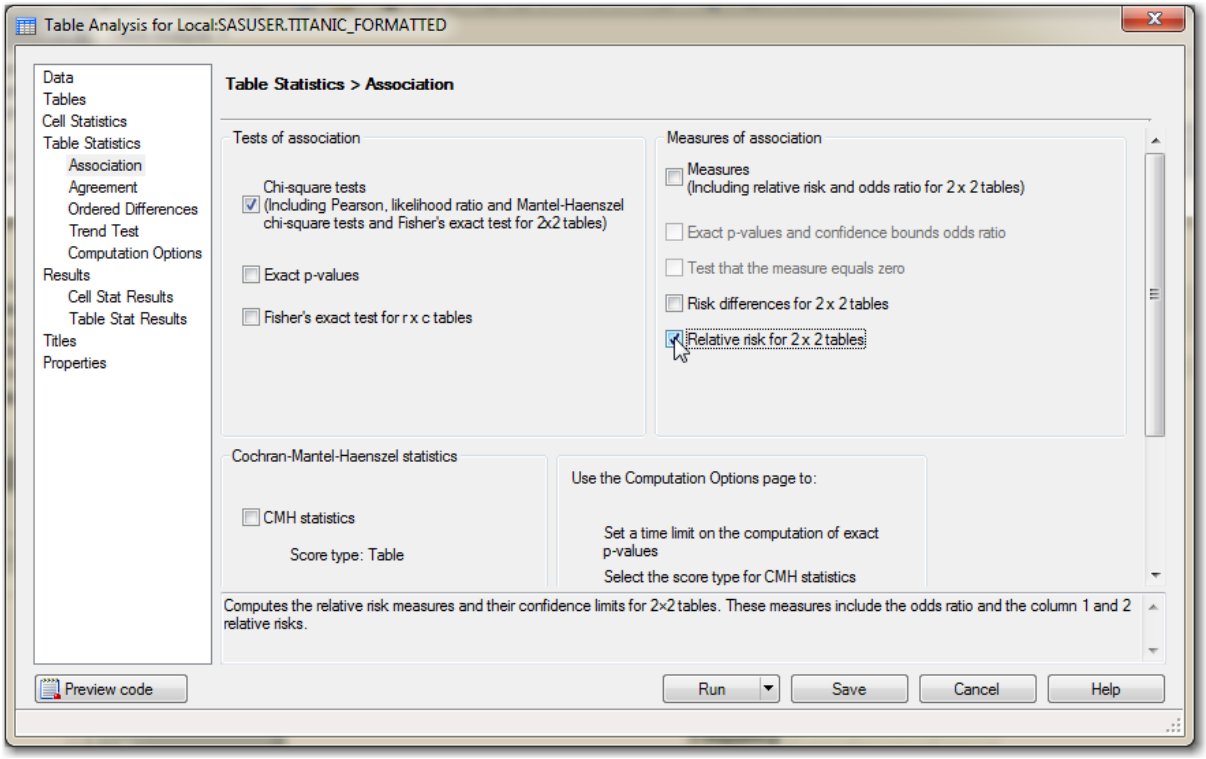


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

3. With **Cell Statistics** selected at the left, check the boxes for **Cell contribution to Pearson chi-square** and **Expected cell frequency**.



4. Select **Association** at the left. Then check the box for **Chi-square tests** under **Tests of association** and **Relative risk for 2×2 tables** under **Measures of association**.



5. Click **Run**. Do not replace the results from the previous run.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Output

Table of Gender by Survival			
Gender	**Survival**		
Frequency Expected Cell Chi-Square Row Pct	**Died**	**Survived**	**Total**
Female	127 288 90.005 27.25	339 178 145.63 72.75	466
Male	682 521 49.753 80.90	161 322 80.501 19.10	843
Total	809	500	1309

It appears that the cell for **Survival**=**`1`** (**`Survived`**) and **Gender**=**`Female`** contributes the most to the chi-square statistic. The Cell Chi-Square value is 145.63.

The cell chi-square is calculated using the formula
(observed frequency – expected frequency)^2 / expected frequency.

The overall chi-square statistic is calculated by adding the cell chi-square values over all rows and columns: $\Sigma\Sigma$ ((observed$_{rc}$ – expected$_{rc}$)^2 / expected$_{rc}$).

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Below is the table that shows the chi-square test and Cramer's V.

Statistic	DF	Value	Prob
Chi-Square	1	365.8869	<.0001
Likelihood Ratio Chi-Square	1	372.9213	<.0001
Continuity Adj. Chi-Square	1	363.6179	<.0001
Mantel-Haenszel Chi-Square	1	365.6074	<.0001
Phi Coefficient		-0.5287	
Contingency Coefficient		0.4674	
Cramer's V		-0.5287	

Because the *p*-value for the chi-square statistic is <.0001, which is below .05, you reject the null hypothesis at the 0.05 level and conclude that there is evidence of an association between **Gender** and **Survival**. Cramer's V of -0.5287 indicates that the association detected with the chi-square test is relatively strong.

Fisher's Exact Test	
Cell (1,1) Frequency (F)	127
Left-sided Pr <= F	7.351E-83
Right-sided Pr >= F	1.0000
Table Probability (P)	6.705E-83
Two-sided Pr <= P	7.918E-83

Exact tests are often useful where asymptotic distributional assumptions are not met. The usual guidelines for the asymptotic chi-square test are generally 20-25 total observations for a 2×2 table, with 80% of the table cells having counts greater than 5. Fisher's Exact Test is provided by the Table Analysis task when tests of association are requested for 2×2 tables. The two-sided *p*-value of $7.918 * 10^{-83}$ is exceedingly small and statistically significant.

Estimates of the Relative Risk (Row1/Row2)			
Type of Study	**Value**	**95% Confidence Limits**	
Case-Control (Odds Ratio)	0.0884	0.0677	0.1155
Cohort (Col1 Risk)	0.3369	0.2894	0.3921
Cohort (Col2 Risk)	3.8090	3.2797	4.4239

The Relative Risk table shows another measure of strength of association.

The odds ratio is shown in the first row of the table, along with the 95% confidence limits. The odds ratio can be interpreted as the odds of a top row (**`female`**, in this case) value to be in the left column (**`Died`**), compared with the same odds in the bottom row (**`male`**). The value of 0.0884 says that a female has about 9% of the odds of dying, compared with a male. This is equivalent to saying that a male has about 9% of the odds of surviving, compared with a female.

Cohort estimates for each column are interpreted as probability ratios, rather than odds ratios. You get a choice of assessing probabilities of the left column (Col1) or the right column (Col2). For example, the Col1 risk shows the ratio of the probabilities of females to males being in the left column (27.25/80.90=0.3369).

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

If is often easier to report odds ratios by first transforming the decimal value to a percent difference value. The formula for doing that is (OR-1)*100. In the example, you have (0.0884 – 1) * 100 = –91.16%. In other words, males have 91.16 percent lower odds of surviving compared with females.

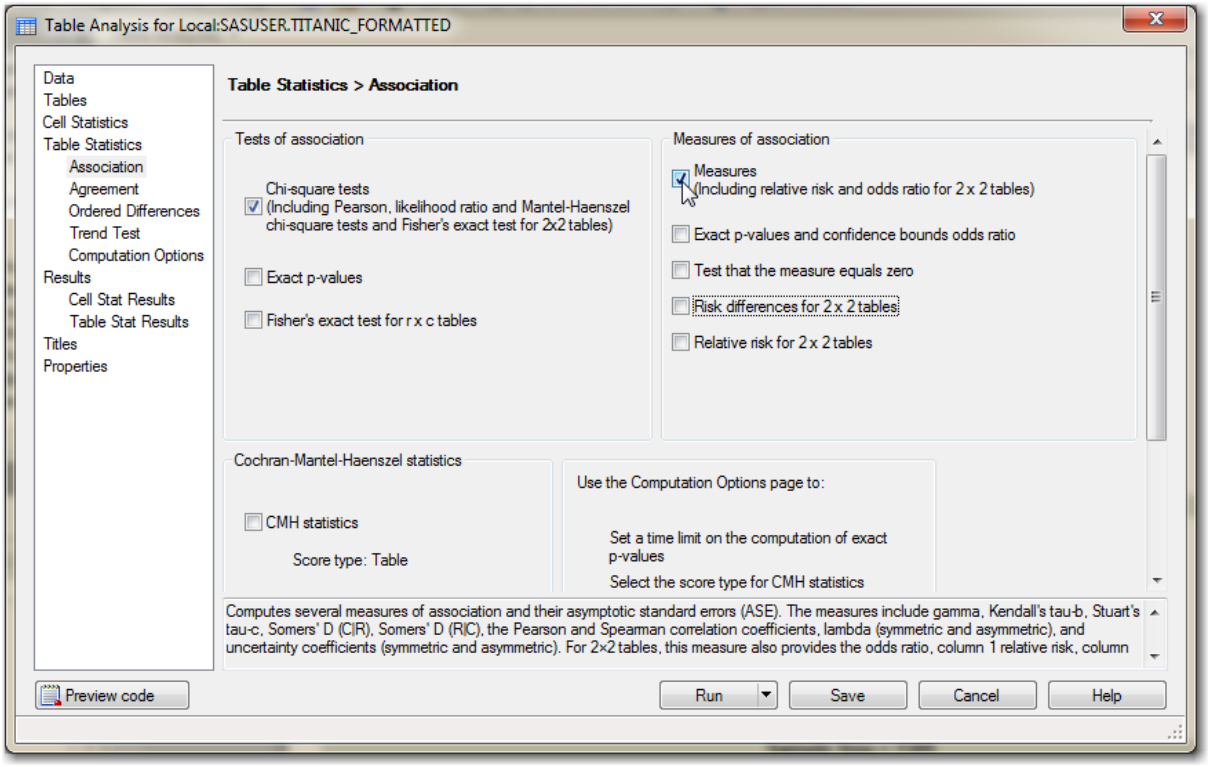
The 95% odds ratio confidence interval goes from 0.0677 to 0.1155. That interval does not include 1. This confirms the statistically significant (at alpha=0.05) result of the Pearson chi-square test of association. A confidence interval that included the value 1 (equality of odds) would be a non-significant result.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Detecting Ordinal Associations

Example: Use the Table Analysis task to test whether an ordinal association exists between **Survival** and **Class**.

1. Reopen the first Table Analysis task by right-clicking it and selecting **Modify** from the menu.
2. With **Tables** selected at the left, remove **Gender** by **Survival** from the **Tables to be generated** list.
3. Select **Association** at the left and check the box for **Chi-square tests** under **Tests of association** and **Measures** under **Measures of Association**.



4. Click **Run**. Do not replace the results from the previous run.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The crosstabulation is shown below.

Table of Class by Survival			
Class(Class of Ticket)	Survival		
Frequency Row Pct	Died	Survived	Total
1st	123 38.08	200 61.92	323
2nd	158 57.04	119 42.96	277
3rd	528 74.47	181 25.53	709
Total	809	500	1309

The results of the Mantel-Haenszel chi-square test are shown below.

Statistic	DF	Value	Prob
Chi-Square	2	127.8592	<.0001
Likelihood Ratio Chi-Square	2	127.7655	<.0001
Mantel-Haenszel Chi-Square	1	127.7093	<.0001
Phi Coefficient		0.3125	
Contingency Coefficient		0.2983	
Cramer's V		0.3125	

Because the *p*-value of the Mantel-Haenszel chi-square is <.0001, you can conclude at the 0.05 significance level that there is evidence of an ordinal association between **Survival** and **Class**.

The Spearman correlation statistic and the 95% confidence bounds are shown below.

Statistic	Value	ASE
Gamma	-0.5067	0.0375
Kendall's Tau-b	-0.2948	0.0253
Stuart's Tau-c	-0.3141	0.0274
Somers' D C|R	-0.2613	0.0226
Somers' D R|C	-0.3326	0.0286
Pearson Correlation	-0.3125	0.0267
Spearman Correlation	-0.3097	0.0266
Lambda Asymmetric C|R	0.1540	0.0331
Lambda Asymmetric R|C	0.0317	0.0320
Lambda Symmetric	0.0873	0.0292
Uncertainty Coefficient C|R	0.0734	0.0127
Uncertainty Coefficient R|C	0.0485	0.0084
Uncertainty Coefficient Symmetric	0.0584	0.0101

The Spearman Correlation (-.3097) indicates that there is a moderate, negative ordinal relationship between **Class** and **Survival** (that is, as **Class** levels increase, **Survival** tends to decrease).

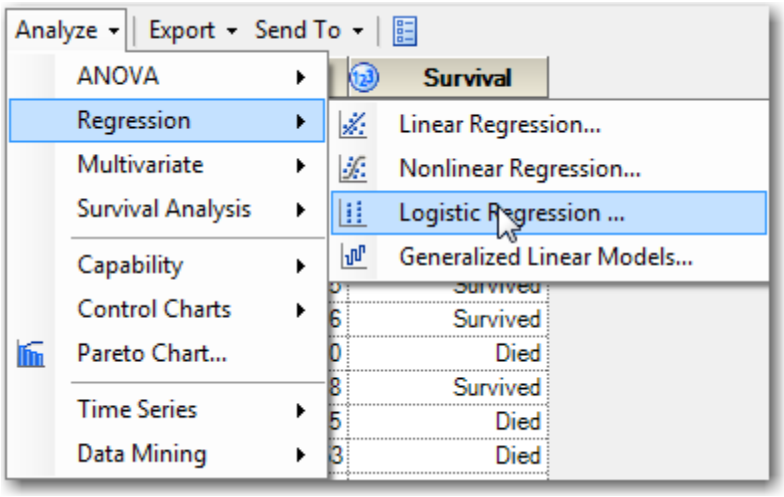
The ASE is the asymptotic standard error (0.0266), which is an appropriate measure of the standard error for larger samples.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Simple Logistic Regression Model

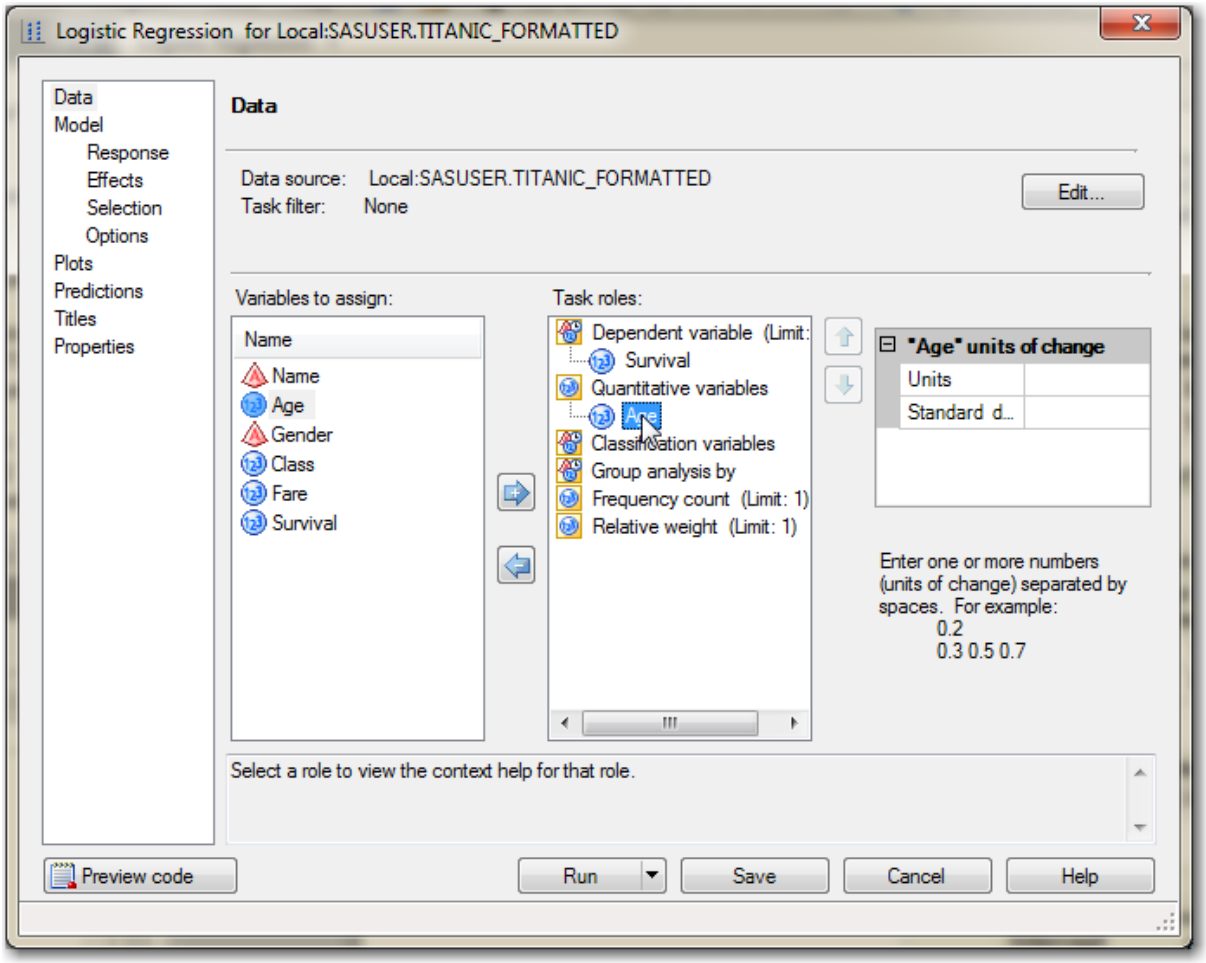
Example: Fit a binary logistic regression model using the Linear Regression task. Select **Survival** as the outcome variable and **Age** as the predictor variable. Model the probability of surviving and request profile likelihood confidence intervals around the estimated odds ratios.

1. With the **Titanic_Formatted** data set active, select **Analyze** ⇨ **Regression** ⇨ **Logistic Regression**.



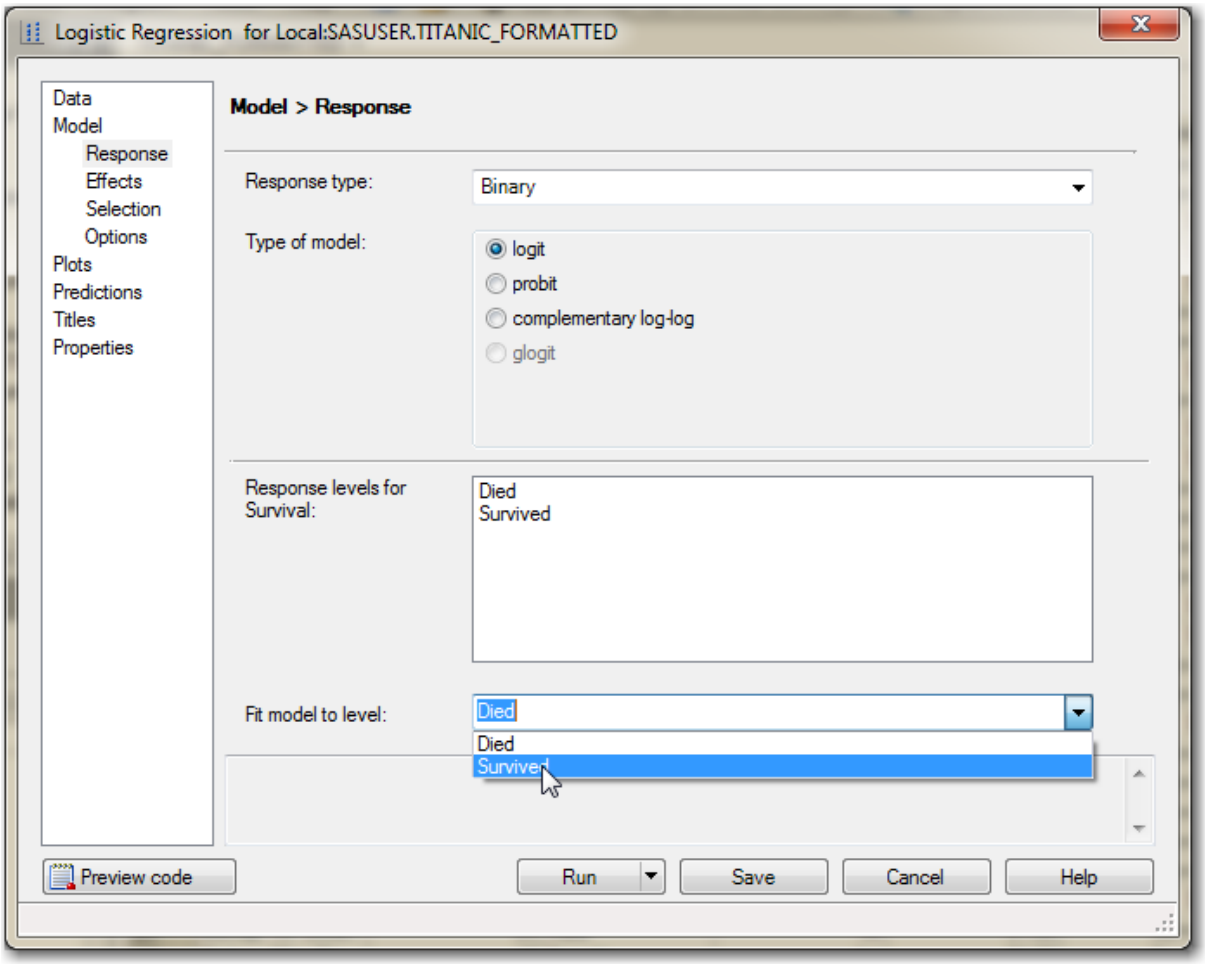
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

2. Assign **Survival** to the dependent variable task role and **Age** to the quantitative variables role.



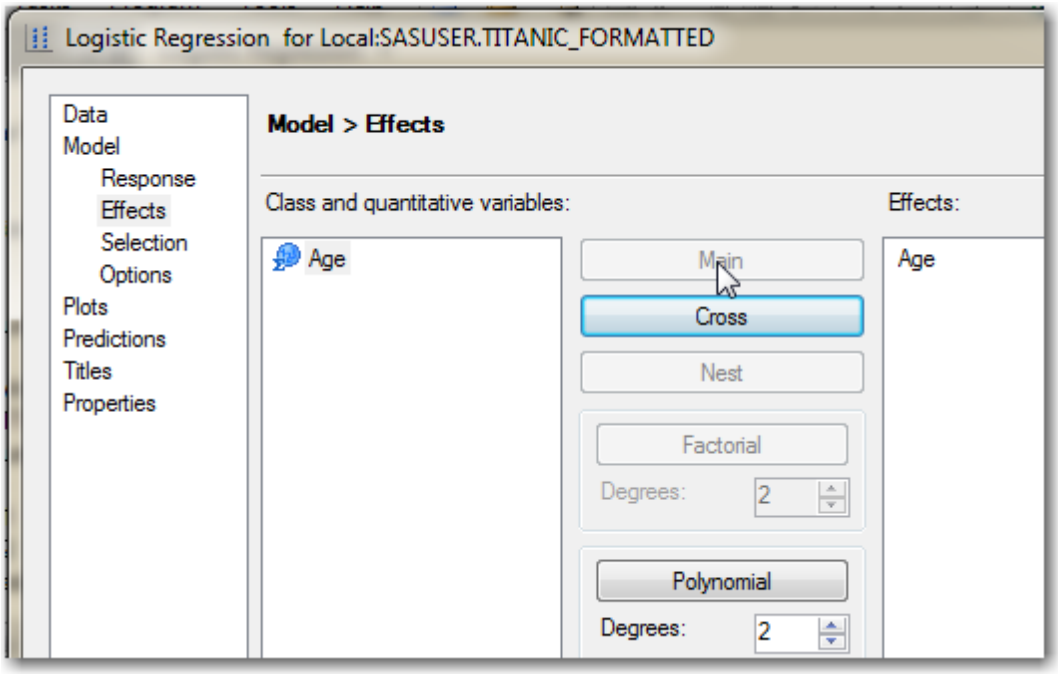
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

3. With **Response** selected at the left, assure that the value of **Fit model to level** is set to **Survived**. Note that **Response levels for Survival** displays the possible values in the data set to fit.



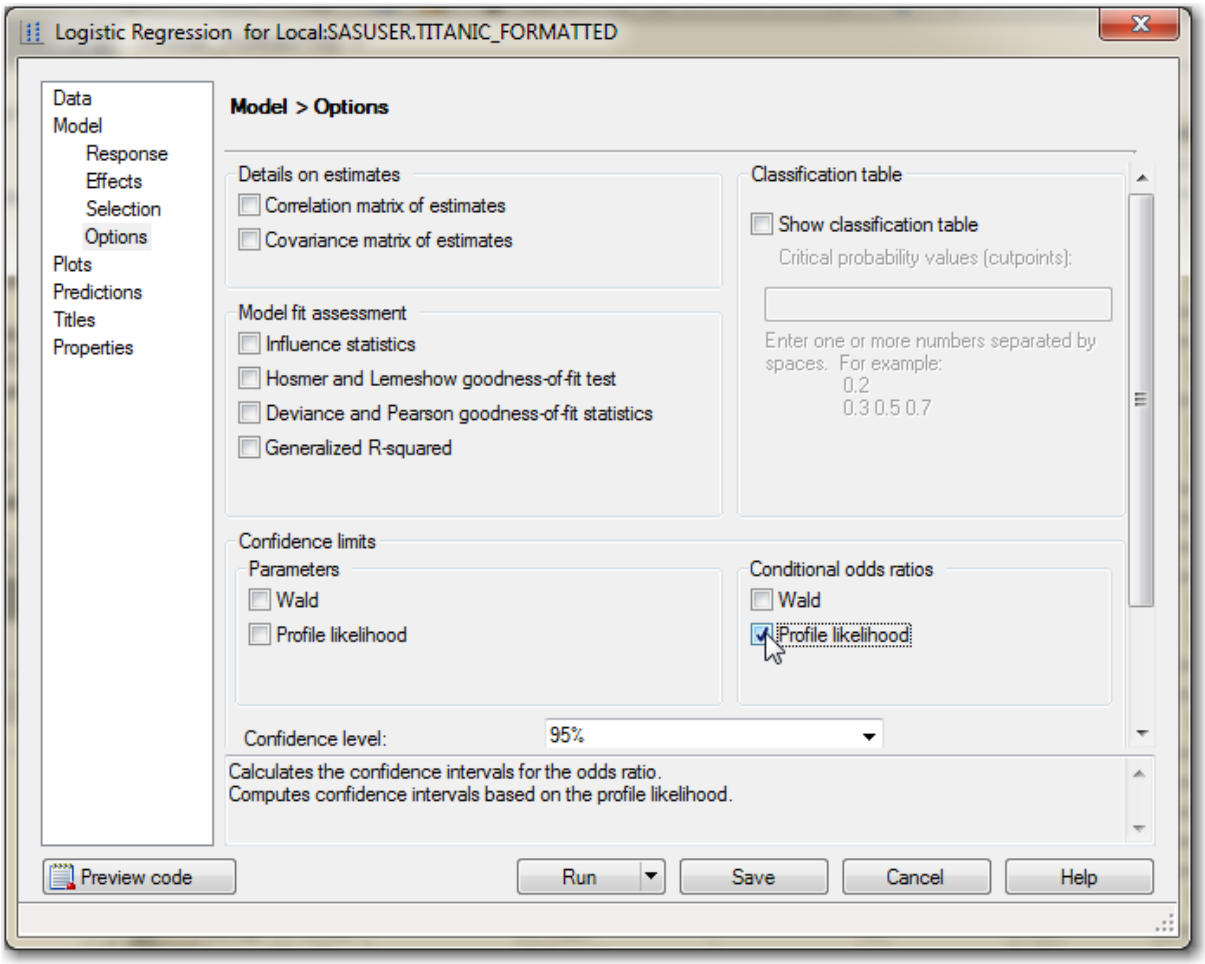
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. With **Effects** selected at the left, add **Age** as a **Main** effect in the model.



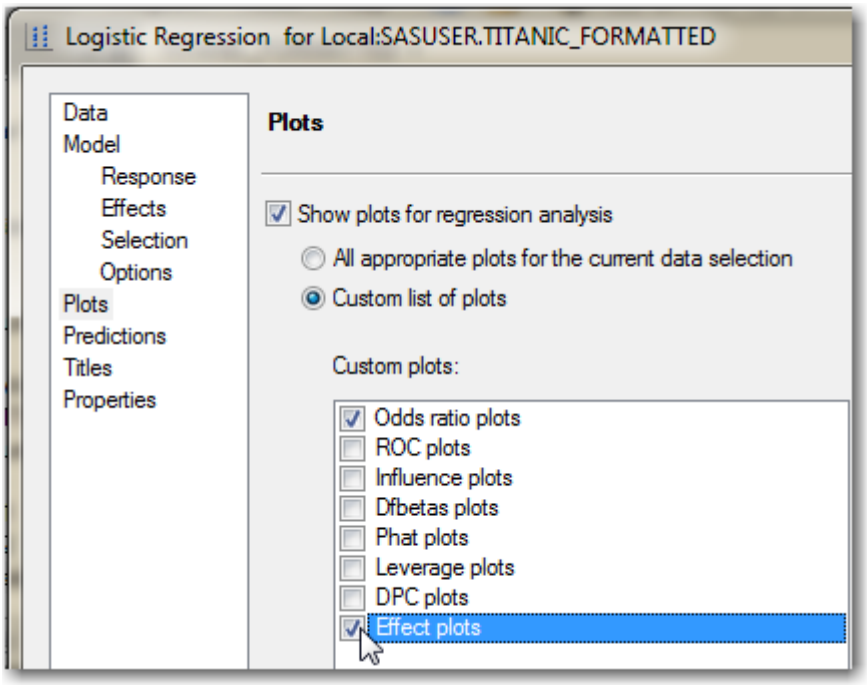
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

5. With **Options** selected at the left, check the box for **Profile likelihood** under **Conditional odds ratios** in the middle of the window.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. With **Plots** selected at the left, select **Custom list of plots**. Then check only the boxes next to **Odds ratio plots** and **Effect plots** in the list.



7. Click **Run**.

Output

Model Information	
Data Set	WORK.SORTTEMPTABLESORTED
Response Variable	Survival
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

The Model Information table describes the data set, the response variable, the number of response levels, the type of model, the algorithm used to obtain the parameter estimates, and the number of observations read and used.

The Optimization Technique is the iterative numerical technique that the Logistic Regression task uses to estimate the model parameters.

The model is assumed to be "binary logit" when there are exactly two response Levels.

Number of Observations Read	1309
Number of Observations Used	1046

The Number of Observations Used is the count of all observations that are nonmissing for all variables specified as effects in the model. The ages of 263 of these 1309 passengers cannot be determined and cannot be used to estimate the model.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Response Profile		
Ordered Value	Survival	Total Frequency
1	Died	619
2	Survived	427

The Response Profile table shows the response variable values listed according to their ordered values. It also shows frequencies of response values.

Probability modeled is Survival='Survived'.

By default, the Linear Regression task orders the response variable alphanumerically and it models the probability of the smallest value. However, you selected **Survived=1** as the response level to model. It is advisable to check that the modeled response level is the one that you intended.

Note: 263 observations were deleted due to missing values for the response or explanatory variables.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

The Model Convergence Status simply informs you that the convergence criterion was met. There are a number of options to control the convergence criterion, but the default is the gradient convergence criterion with a default value of 1E-8 (0.00000001).

The optimization technique does not always converge to a maximum likelihood solution. When this is the case, the output after this point cannot be trusted. Always check to see that the Convergence criterion is satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1416.620	1415.301
SC	1421.573	1425.207
-2 Log L	1414.620	1411.301

The Model Fit Statistics provides three tests:

- AIC is Akaike's 'A' information criterion.
- SC is the Schwarz criterion.
- –2 Log L is –2 times the natural log of the likelihood.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

-2 Log L, AIC, and SC are goodness-of-fit measures that you can use to compare one model to another. ***These statistics measure relative fit among models, but they do not measure absolute fit of any single model.*** Smaller values for all of these measures indicate better fit. However, -2 Log L can be reduced by simply adding more regression parameters to the model. Therefore, it is not used to compare the fit of models that use different numbers of parameters. AIC adjusts for the number of predictor variables, and SCs adjust for the number of predictor variables and the number of observations. SC uses a bigger penalty for extra variables and therefore favors more parsimonious models.

Testing Global Null Hypothesis: BETA=0			
Test	**Chi-Square**	**DF**	**Pr > ChiSq**
Likelihood Ratio	3.3191	1	0.0685
Score	3.3041	1	0.0691
Wald	3.2932	1	0.0696

The Testing Global Null Hypothesis: BETA=0 table provides three statistics to test the null hypothesis that all regression coefficients of the model are 0.

A significant *p*-value for these tests provides evidence that at least one of the regression coefficients for an explanatory variable is not zero. In this way, they are similar to the overall *F* test in linear regression. The Likelihood Ratio Chi-Square is calculated as the difference between the -2 Log L value of the baseline model (Intercept Only) and the -2 Log L value of the hypothesized model (Intercept and Covariates). The statistic is a distributed asymptotically chi-square with degrees of freedom equal to the difference in number of parameters between the hypothesized model and the baseline model. The Score and Wald tests are also used to test whether all the regression coefficients are 0. The likelihood ratio test is the most reliable, especially for small sample sizes (Agresti 1996). All three tests are asymptotically equivalent and often give very similar values.

Wald statistics (*p*-values and confidence limits) require fewer computations to perform and are therefore the default for most output in the Logistic Regression task.

Analysis of Maximum Likelihood Estimates					
Parameter	**DF**	**Estimate**	**Standard Error**	**Wald Chi-Square**	**Pr > ChiSq**
Intercept	1	-0.1335	0.1448	0.8501	0.3565
Age	1	-0.00800	0.00441	3.2932	0.0696

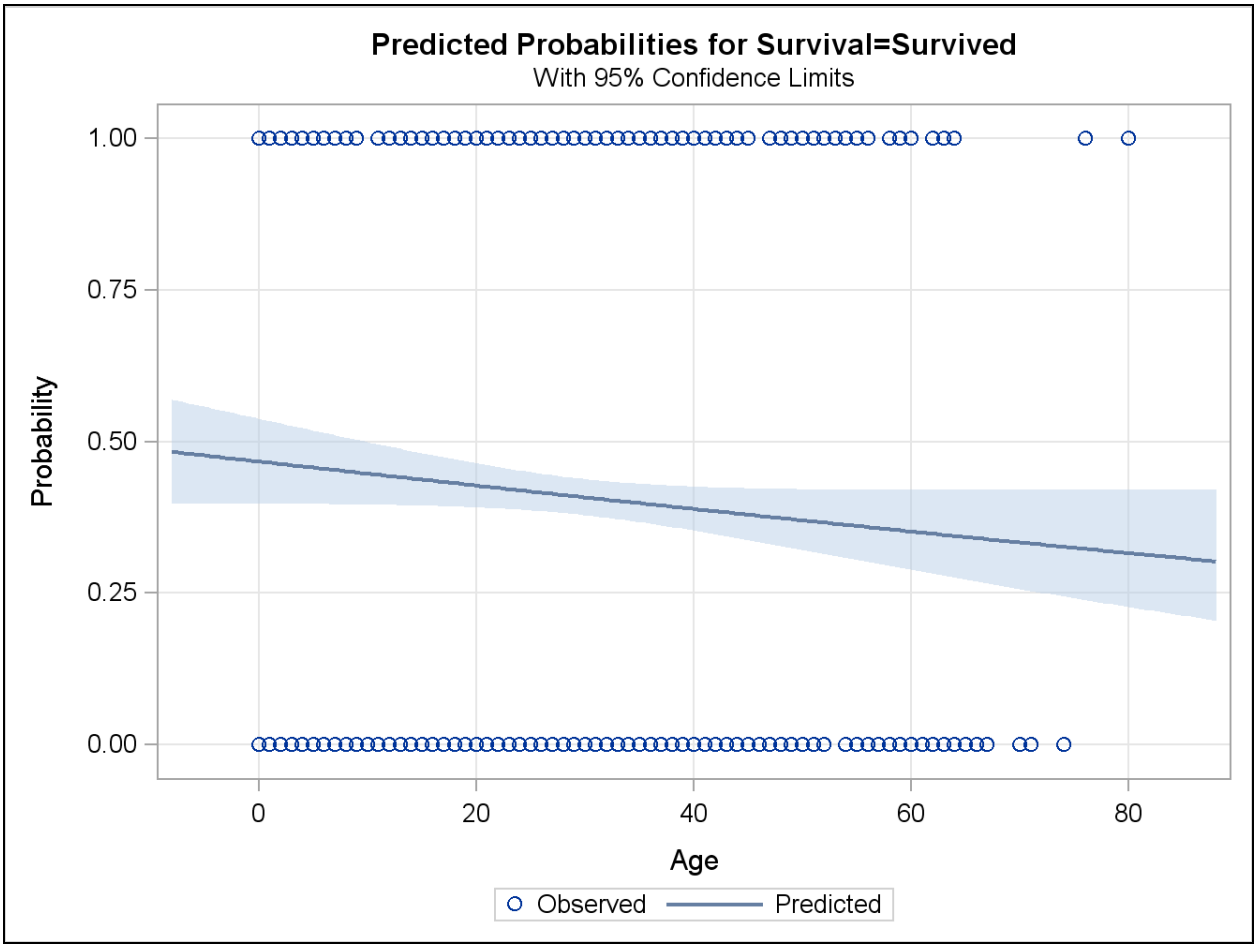
The Analysis of Maximum Likelihood Estimates table lists the estimated model parameters, their standard errors, Wald Chi-Square values, and *p*-values.

The parameter estimates are the estimated coefficients of the fitted logistic regression model. The logistic regression equation is logit($\hat{p}$) = –0.1335 + (- 0.00800) * **Age** for this example.

The Wald chi-square and its associated *p*-value tests whether the parameter estimate is significantly different from 0. For this example, the *p*-values for the variable **Age** is not significant at the 0.05 significance level (*p*-value=0.0696).

Even though **Age** is not a statistically significant predictor in a simple logistic model, it might still be significant when included as a predictor in a multivariate model.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

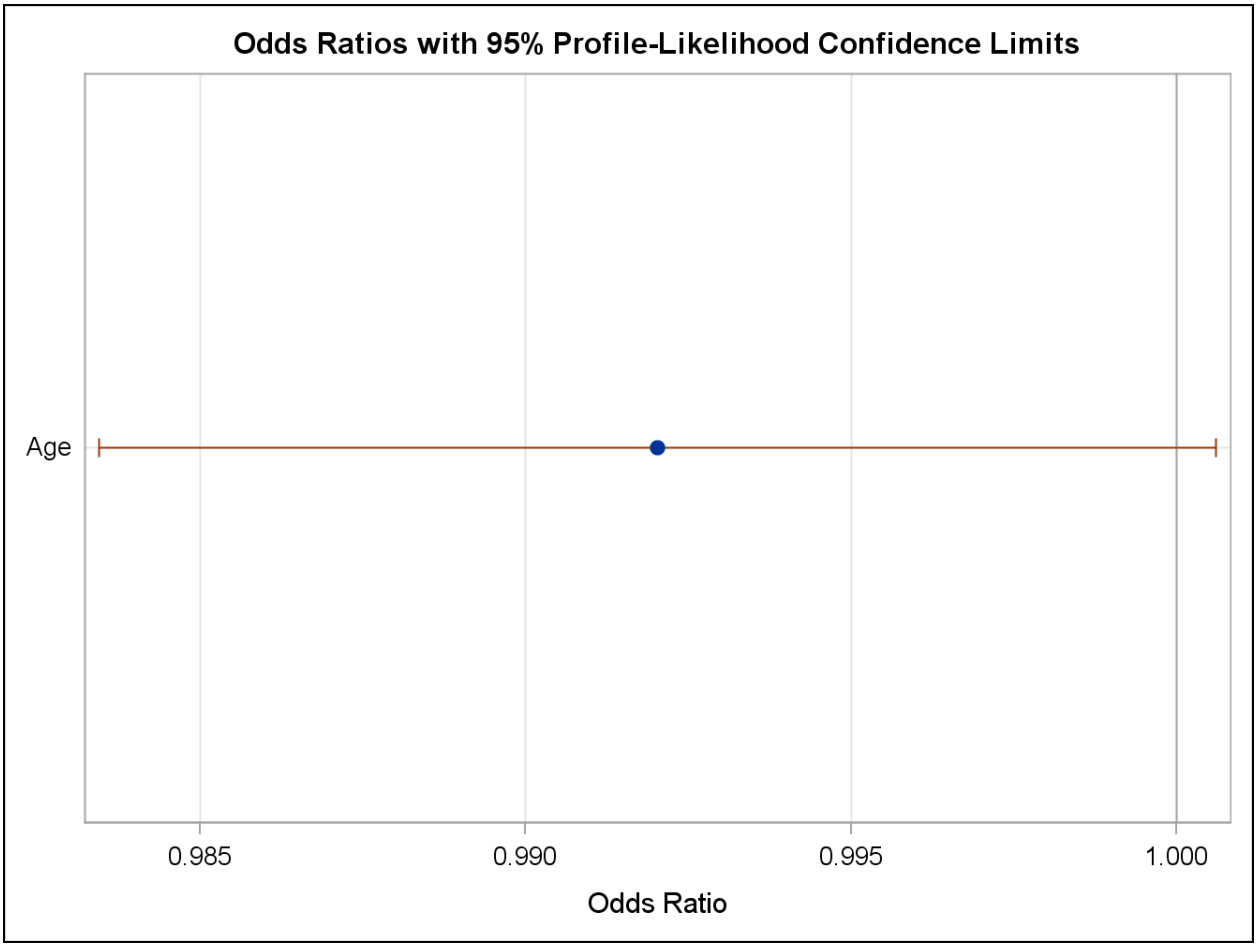


The estimated model is displayed on the probability scale in the Effect Plot. The observed values are plotted at probabilities 1.00 and 0.00.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	51.3	**Somers' D**	0.050
Percent Discordant	46.4	**Gamma**	0.051
Percent Tied	2.3	**Tau-a**	0.024
Pairs	264313	**c**	0.525

Odds Ratio Estimates and Profile-Likelihood Confidence Intervals				
Effect	**Unit**	**Estimate**	**95% Confidence Limits**	
Age	1.0000	0.992	0.983	1.001



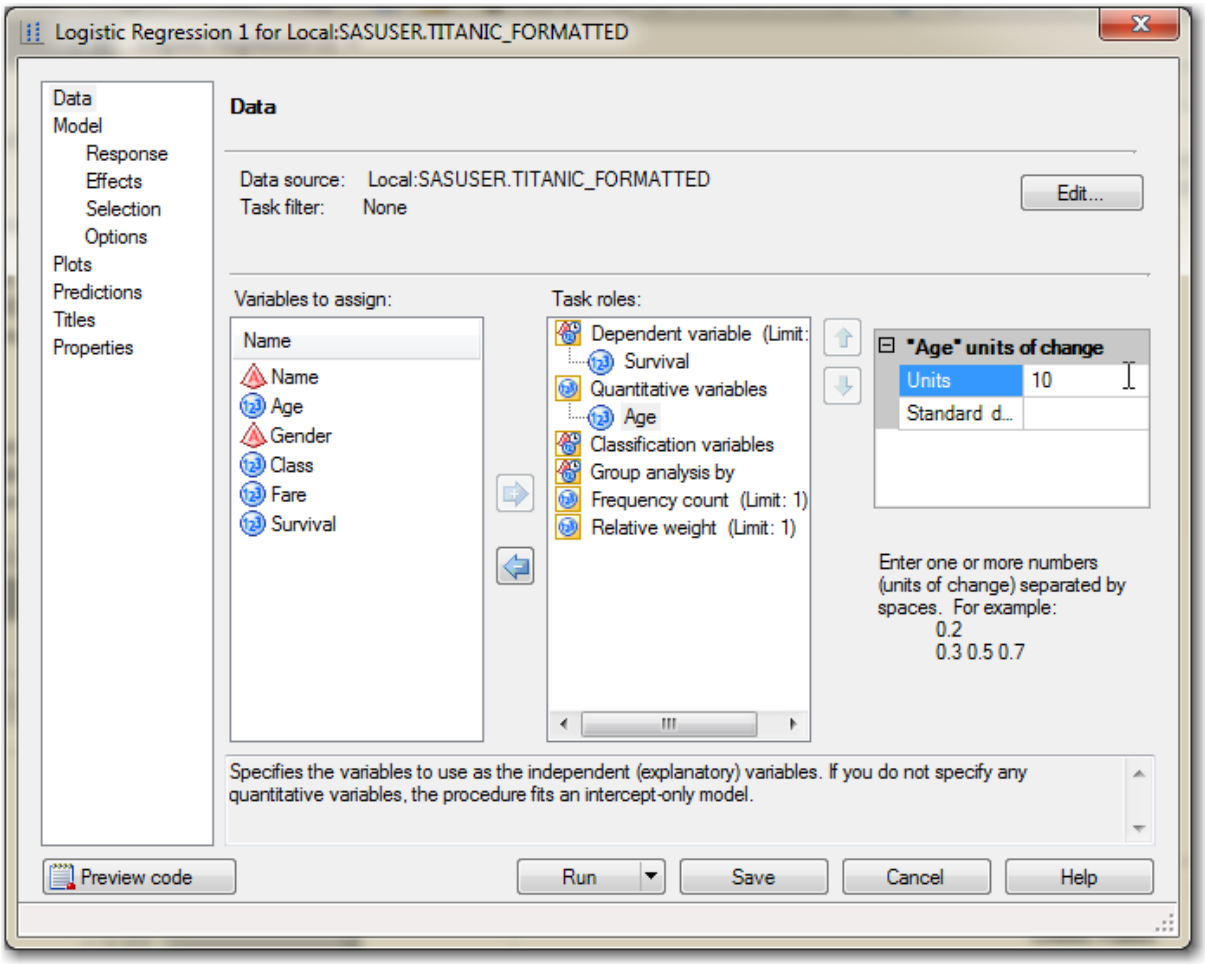
The above tables and plots are described in detail in the next slides.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Multiple Logistic Regression with Categorical Predictors

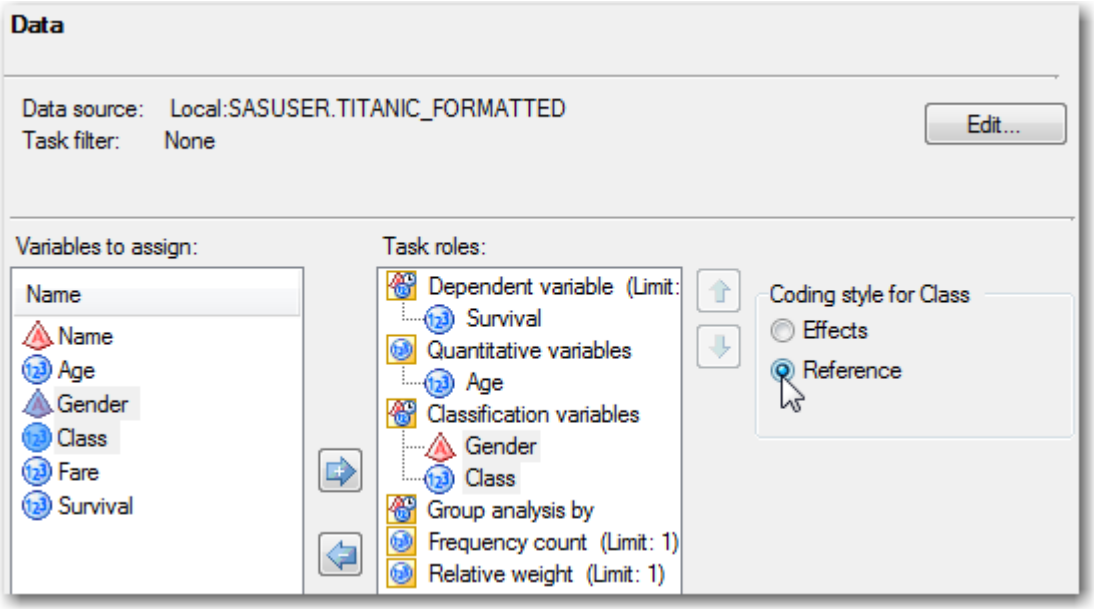
Example: Fit a binary logistic regression model with the Logistic Regression task. Select **Survival** as the outcome variable and **Age**, **Gender**, and **Class** as the predictor variables. Request profile likelihood confidence intervals around the estimated odds ratios.

1. Modify the previous Logistic Regression task.
2. With **Data** selected at right, highlight **Age**.
3. In the extreme right table. **"Age" units of change**, click in the space to the right of **Units**. Type **10** in the space. This enables you to analyze the odds ratio effects for a 10-year difference in age (rather than the default of one year).

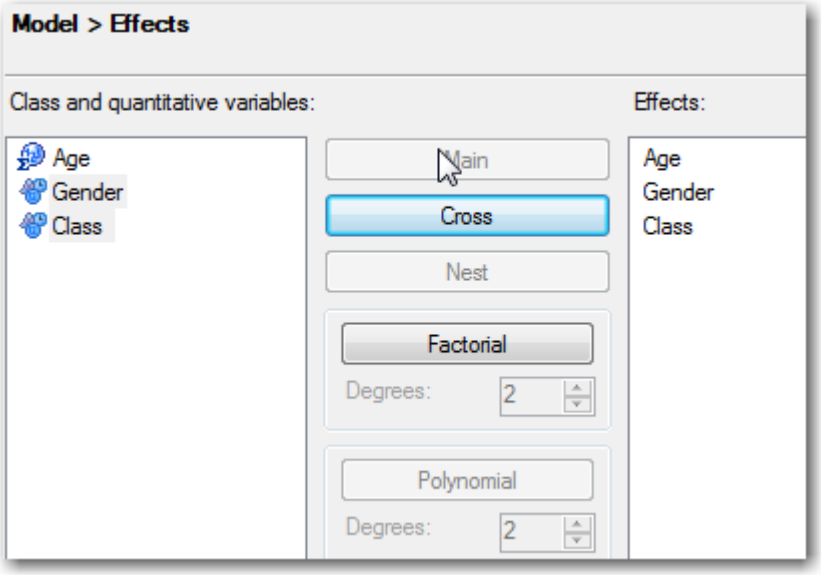


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. Add **Gender** and **Class** to the classification variables role and select **Reference** as the coding style for **Gender** and also for **Class**. **Male** will be the default reference level because it is the highest in alphanumeric order and **3rd Class** will be the reference level for **Class**.



5. With **Effects** selected at the left, add **Gender** and **Class** as **Main** effects in the model.



6. Click **Run**. Do not replace the results from the previous run.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Output

Model Information	
Data Set	WORK.SORTTEMPTABLESORTED
Response Variable	Survival
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1309
Number of Observations Used	1046

Response Profile		
Ordered Value	**Survival**	**Total Frequency**
1	Died	619
2	Survived	427

Probability modeled is Survival='Survived'.

Class Level Information			
Class	**Value**	**Design Variables**	
Gender	**Female**	1	
	Male	0	
Class	**1st**	1	0
	2nd	0	1
	3rd	0	0

The Class Level Information table includes the predictor variables chosen as Classification variable. Because you used reference coding, this table reflects your choice of **Male** as the reference level. The design variable is 1 when **Gender** is **Female** and 0 when **Gender** is **Male**. The reference level for **Class** is **3rd**, so there or two design variables, each coded 0 for observations where **Class** is**3rd**.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1416.620	992.315
SC	1421.573	1017.079
-2 Log L	1414.620	982.315

The SC value in the **Age** only model was 1425.207. Here it is 1017.079. Recalling that smaller values imply better fit, you can conclude that this model is better fitting.

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	432.3052	4	<.0001
Score	386.1522	4	<.0001
Wald	277.3202	4	<.0001

This model is statistically significant; indicating at least one of the predictors in the model is useful in predicting survival.

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Age	1	29.6314	<.0001
Gender	1	226.2235	<.0001
Class	2	103.3575	<.0001

The Type 3 Analysis of Effects table is generated when a predictor variable is defined as a classification variable. This analysis is similar to the individual tests in the Linear Models task parameter estimates table. Just as in ANOVA and Linear Regression, these are adjusted effects.

All effects, including the **Age** effect, which was not statistically significant in the univariate model, are statistically significant.

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-1.2628	0.2030	38.7108	<.0001
Age		1	-0.0345	0.00633	29.6314	<.0001
Gender	Female	1	2.4976	0.1661	226.2235	<.0001
Class	1st	1	2.2907	0.2258	102.8824	<.0001
Class	2nd	1	1.0093	0.1984	25.8849	<.0001

For CLASS variables, effects are displayed for each of the design variables. Because Reference Cell coding was used, each effect is measured against the reference level. For example, the estimate for **Gender | female** shows the difference in logits between females and males. **Class | 1** shows the logit difference between first-class passengers and third-class passengers and **Class | 2** shows the difference in logits between second class and third class. All of these contrasts are statistically significant.

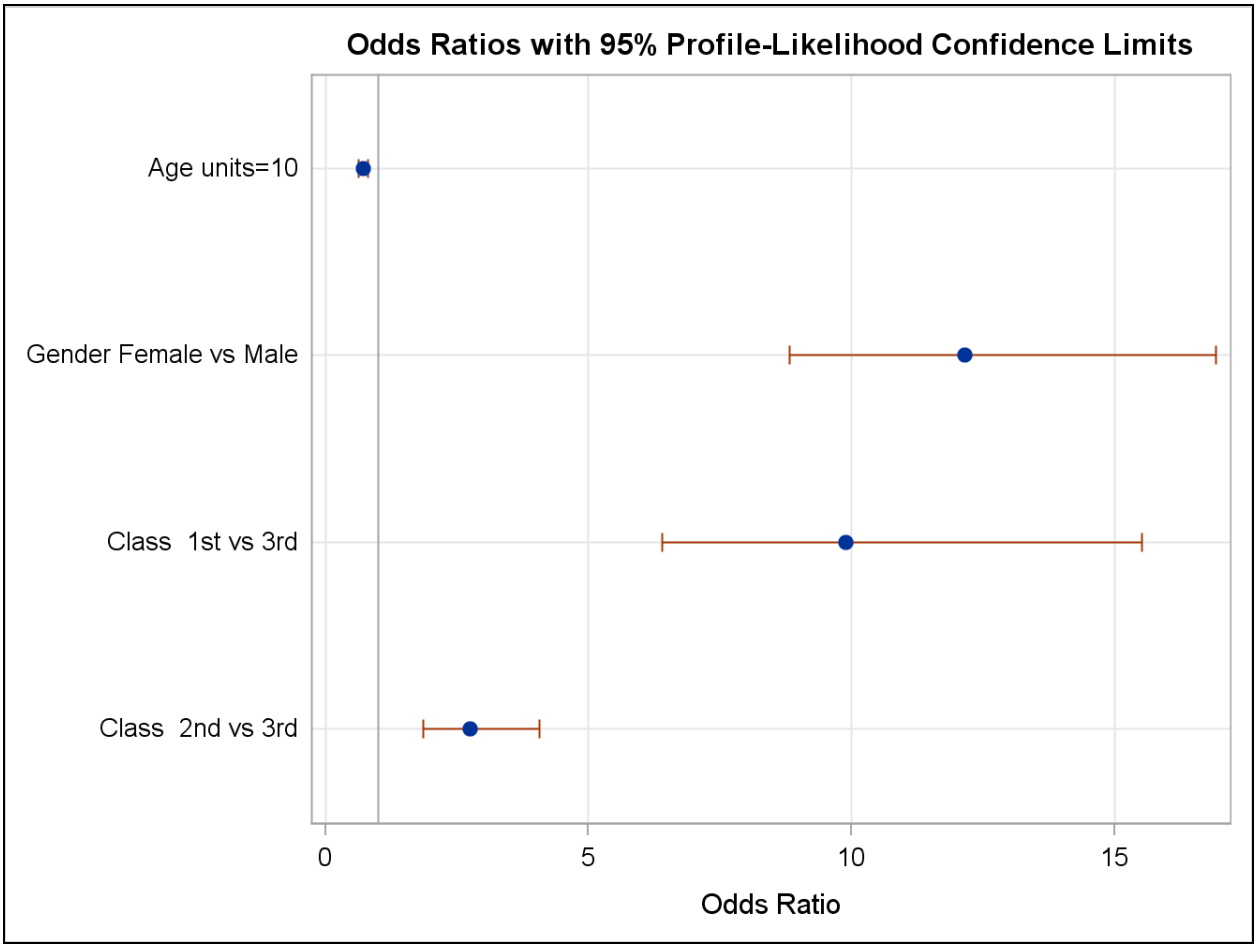
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	83.8	Somers' D	0.680
Percent Discordant	15.8	Gamma	0.683
Percent Tied	0.4	Tau-a	0.329
Pairs	264313	c	0.840

The c (Concordance) statistic value is 0.840 for this model, indicating that 84% of the positive and negative response pairs are correctly sorted using **Age**, **Gender**, and **Class**.

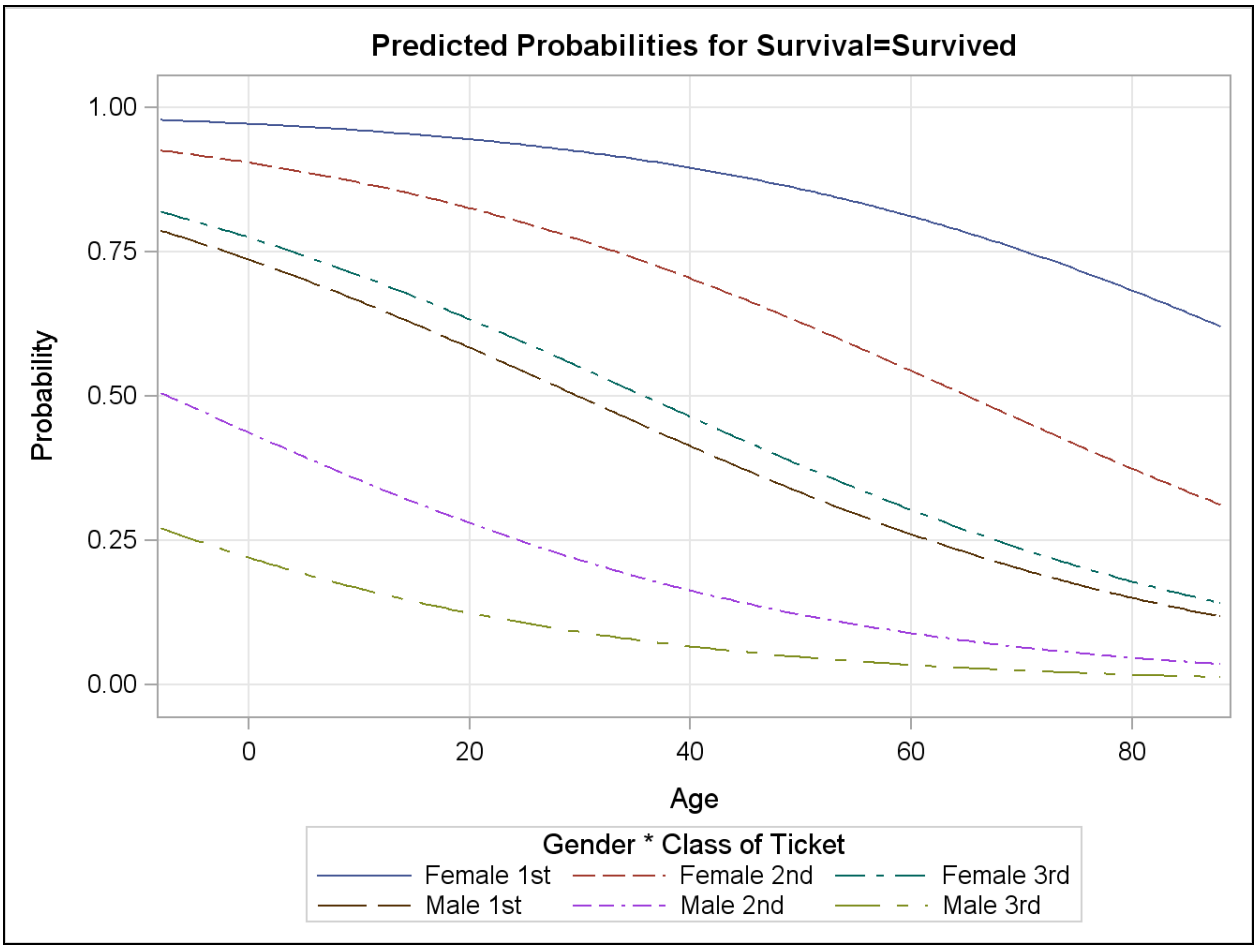
Odds Ratio Estimates and Profile-Likelihood Confidence Intervals				
Effect	Unit	Estimate	95% Confidence Limits	
Age	10.0000	0.708	0.625	0.801
Gender Female vs Male	1.0000	12.153	8.823	16.925
Class 1st vs 3rd	1.0000	9.882	6.395	15.513
Class 2nd vs 3rd	1.0000	2.744	1.863	4.059

The units option that you checked applies to the odds ratio table requested by the Profile likelihood Conditional odds ratio option. The table shows that a 10-year-older age is associated with a 29.2% decrease in survival odds. The plot displays these values graphically.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Finally, the Effects Plot shows the probability of survival across all combinations of categories and levels of all three predictor variables.



This plot applies the parameter estimates from the logistic model to theoretical values of the predictors and then converts the predictions to the probability scale.

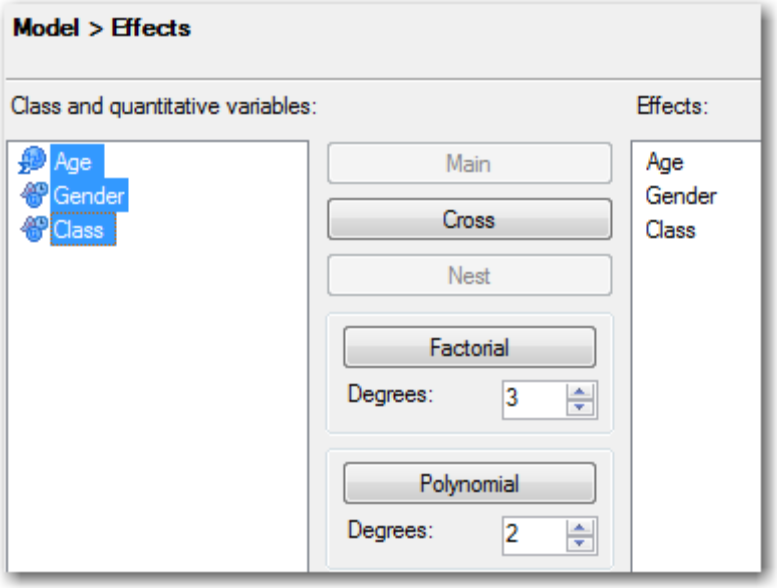
The plot indicates that at every age, all classes of women are predicted to survive at a greater rate than all classes of men. Holding **Age** and **Gender** constant, the descending order of predicted survival probabilities are first class, second class, and third class.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Logistic Regression Backward Elimination with Interactions

Example: Fit a multiple logistic regression model using the backward elimination method. The full model should include all the main effects and two-way interactions.

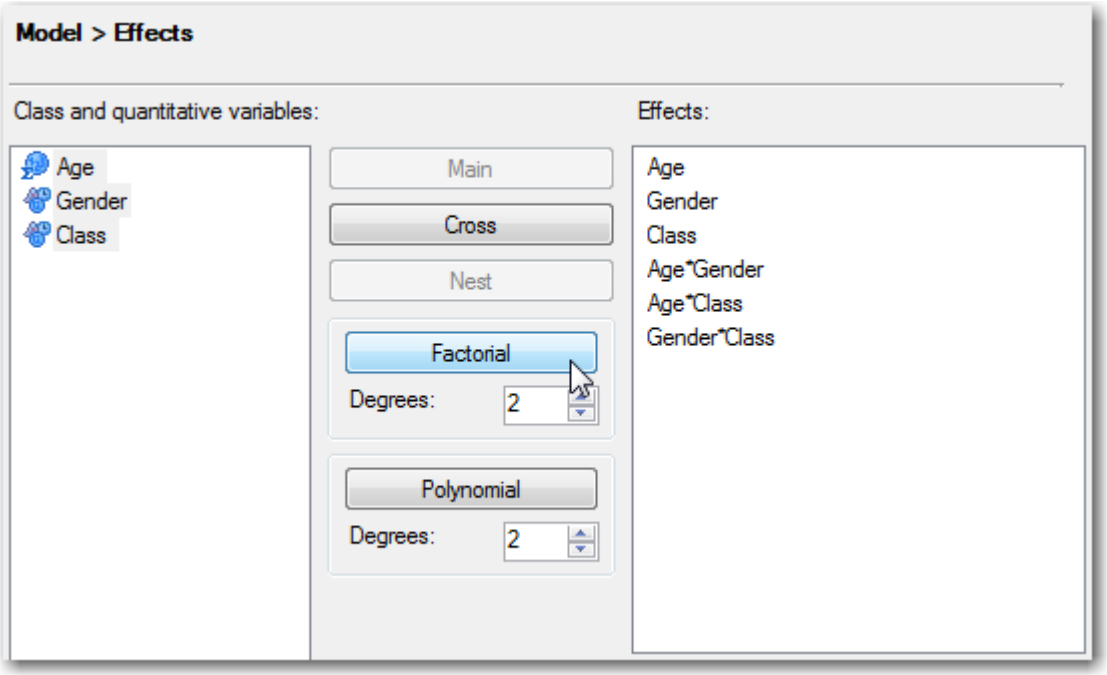
1. Reopen the previous Logistic Regression task window by right-clicking it and selecting **Modify** from the drop-down menu.
2. Select **Effects** under **Model** at left.
3. Highlight **Age**, **Gender**, and **Class** under **Class and quantitative variables** (left-click **Age**, hold the SHIFT key, and then left-click **Class**).



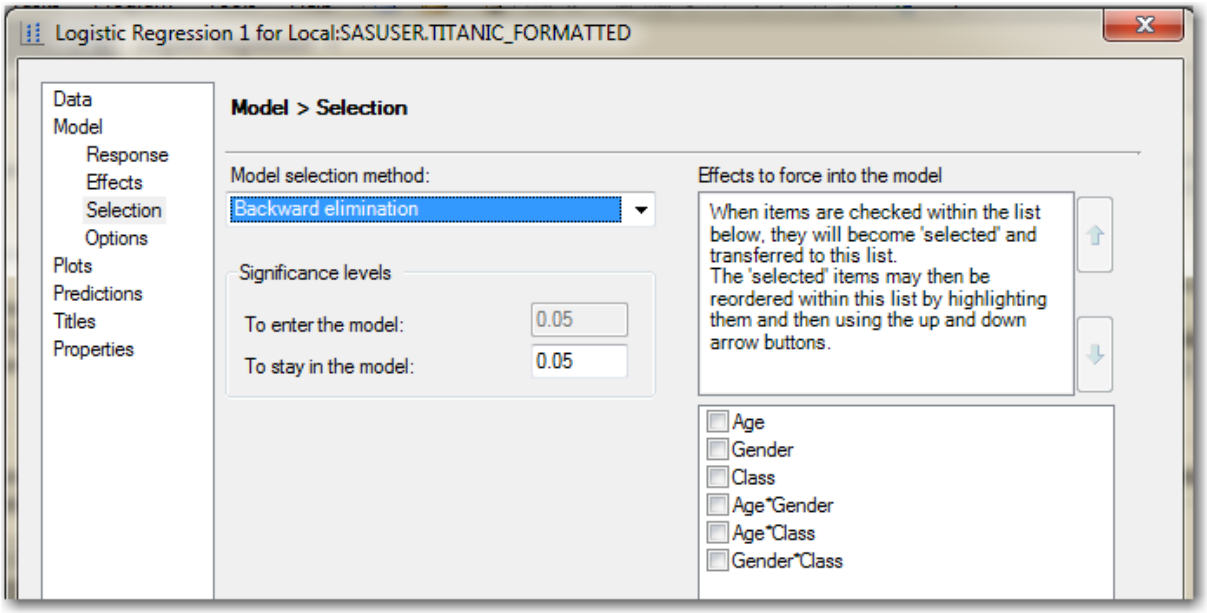
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. Set the value of **Degrees** under **Factorial** to **2** and then click **Factorial**.

If the degrees were set to 3, then the 3-way interaction would be tested, too. If there is a theoretical basis for testing that interaction, then this change would not be made.

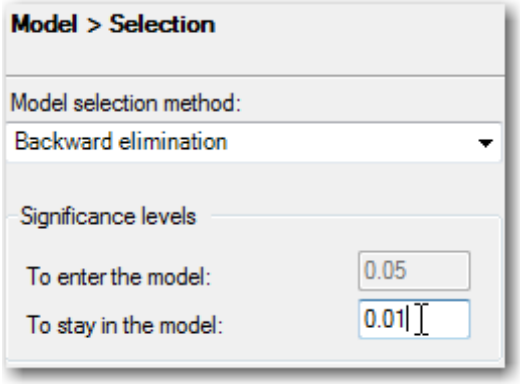


5. Select **Selection** under **Model** at left and then change the **Model selection method** to **Backward** elimination by clicking the down arrow and clicking that selection in the menu.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. Change the significance level **To stay in the model** to **0.01**.



A more conservative significance level criterion was used because the sample size was relatively large and the interactions were not hypothesized a priori. Under these circumstances, there is an elevated risk of creating a model with effects that are significant only by chance. Lowering the significance criteria provides some protection.

7. Click **Run**. Do not replace the results from the previous run.

Output

Model Information	
Data Set	WORK.SORTTEMPTABLESORTED
Response Variable	Survival
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1309
Number of Observations Used	1046

Response Profile		
Ordered Value	**Survival**	**Total Frequency**
1	Died	619
2	Survived	427

Probability modeled is Survival='Survived'.

Note: 263 observations were deleted due to missing values for the response or explanatory variables.

All information to this point is the same as that from the previous model.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Backward Elimination Procedure

Class Level Information			
Class	Value	Design Variables	
Gender	Female	1	
	Male	0	
Class	1st	1	0
	2nd	0	1
	3rd	0	0

The Model Fit Statistics and Testing Global Null Hypothesis tables at Step 0 are presented.

Step 0. The following effects were entered:

Intercept Age Gender Age*Gender Class Age*Class Gender*Class

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1416.620	937.714
SC	1421.573	987.241
-2 Log L	1414.620	917.714

Step 1. Effect Age*Gender is removed:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1416.620	940.064
SC	1421.573	984.638
-2 Log L	1414.620	922.064

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Testing Global Null Hypothesis: BETA=0			
Test	**Chi-Square**	**DF**	**Pr > ChiSq**
Likelihood Ratio	492.5567	8	<.0001
Score	424.3790	8	<.0001
Wald	221.7387	8	<.0001

Residual Chi-Square Test		
Chi-Square	**DF**	**Pr > ChiSq**
4.3665	1	0.0367

Step 2. Effect Age*Class is removed:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	**Intercept Only**	**Intercept and Covariates**
AIC	1416.620	945.832
SC	1421.573	980.501
-2 Log L	1414.620	931.832

Testing Global Null Hypothesis: BETA=0			
Test	**Chi-Square**	**DF**	**Pr > ChiSq**
Likelihood Ratio	482.7886	6	<.0001
Score	422.4668	6	<.0001
Wald	237.1963	6	<.0001

Residual Chi-Square Test		
Chi-Square	**DF**	**Pr > ChiSq**
13.2931	3	0.0040

No (additional) effects met the 0.01 significance level for removal from the model.

The process stops after the two interactions involving **Age** are removed.

Summary of Backward Elimination					
Step	**Effect Removed**	**DF**	**Number In**	**Wald Chi-Square**	**Pr > ChiSq**
1	**Age*Gender**	1	5	4.3264	0.0375
2	**Age*Class**	2	4	8.8477	0.0120

The *p*-values associated with the two removed effects were below 0.05 and would therefore be retained, if the default SLSTAY criterion were used.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Age	1	32.5663	<.0001
Gender	1	40.0553	<.0001
Class	2	44.4898	<.0001
Gender*Class	2	43.9289	<.0001

Analysis of Maximum Likelihood Estimates							
Parameter			DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept			1	-0.6552	0.2113	9.6165	0.0019
Age			1	-0.0385	0.00674	32.5663	<.0001
Gender	Female		1	1.3970	0.2207	40.0553	<.0001
Class	1st		1	1.5770	0.2525	38.9980	<.0001
Class	2nd		1	-0.0242	0.2720	0.0079	0.9292
Gender*Class	Female	1st	1	2.4894	0.5403	21.2279	<.0001
Gender*Class	Female	2nd	1	2.5599	0.4562	31.4930	<.0001

Notice that when an intercept effect is created involving one or more categorical variables, new rows are added to the parameter estimates table. These represent design variables that SAS creates in order to test the interactions.

An interaction between two variables means that the effect of one variable is different at different values of the other variable. This makes the model more complex and difficult to interpret.

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	85.0	Somers' D	0.703
Percent Discordant	14.7	Gamma	0.706
Percent Tied	0.4	Tau-a	0.340
Pairs	264313	c	0.852

The c value is a slight improvement over the previous model (c=0.840) that only included the main effects.

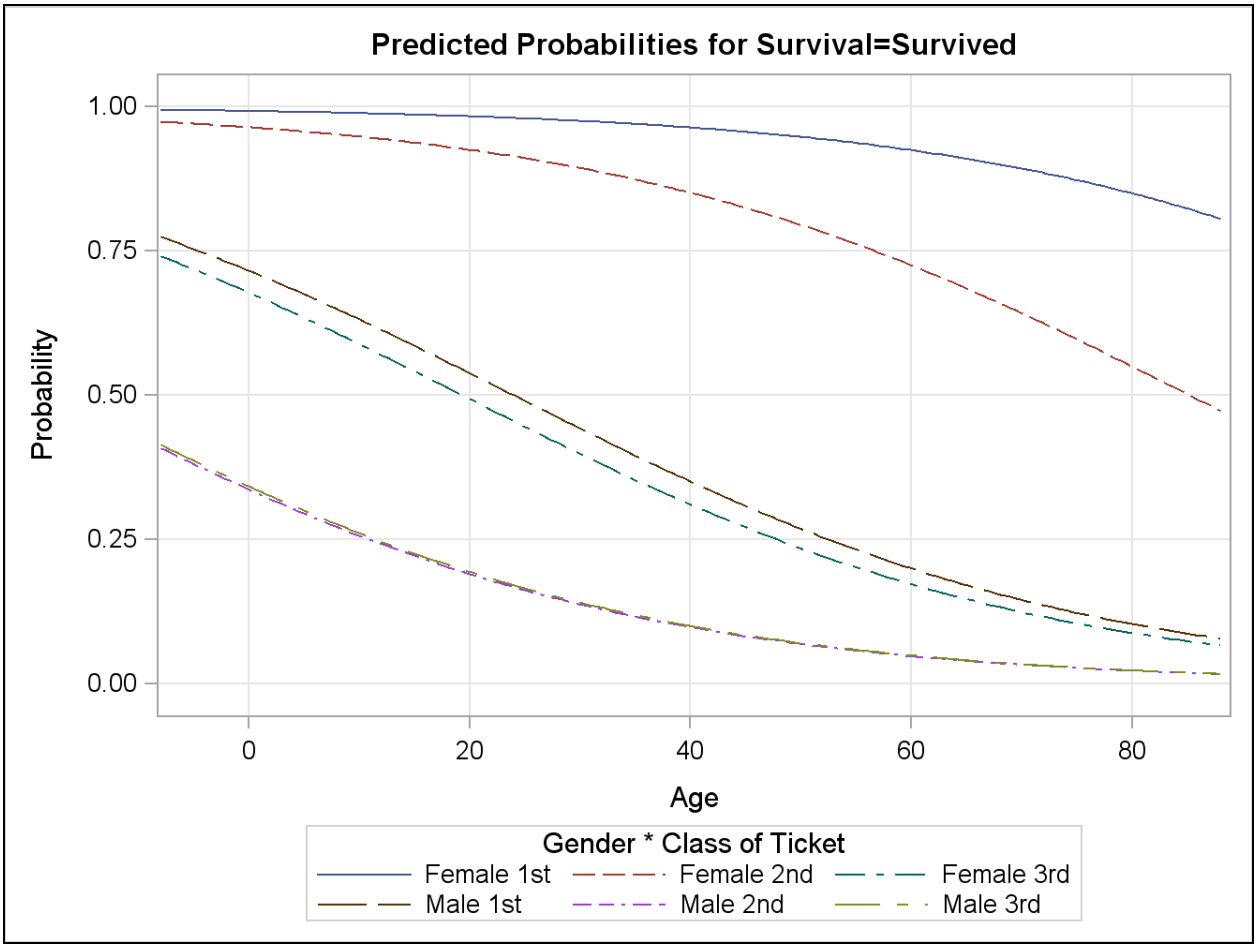
Odds Ratio Estimates and Profile-Likelihood Confidence Intervals				
Effect	Unit	Estimate	95% Confidence Limits	
Age	10.0000	0.681	0.595	0.775

Odds ratios are not calculated for effects involved in interactions. Any single odds ratio for **Class** or for **Gender** would be misleading because the effects vary for each at different levels of the other variable.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The odds ratio for **Age** in this model changed only slightly with the addition of the interaction term.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

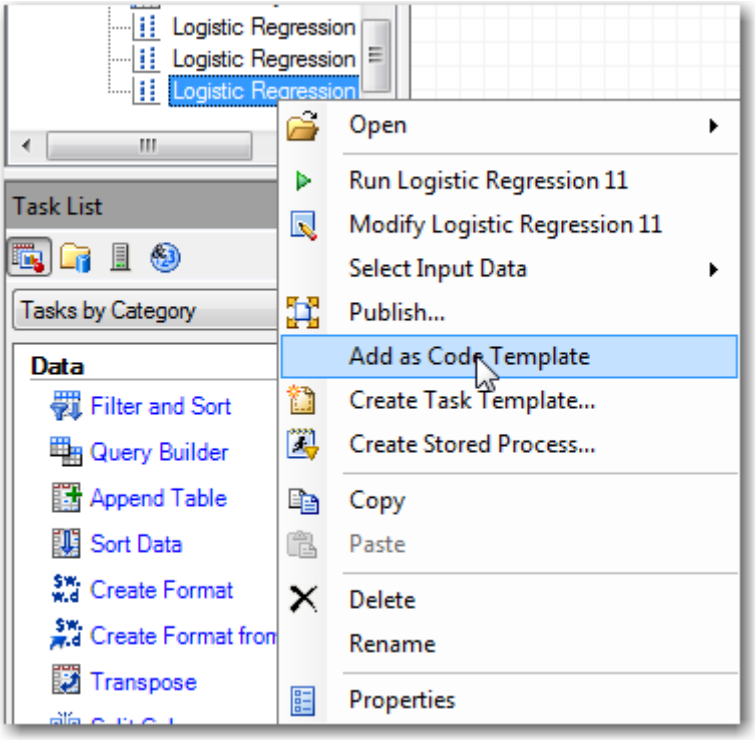


The effect plot shows the interaction. **`Male 2nd`** class is closer to **`Male 3rd`** class (they have almost identical probabilities at every age) than **`Male 1st`** class. **`Female 2nd`** class is much closer to **`Female 1st`** class than to **`Female 3rd`** class.

Self Study: In order to estimate and plot odds ratios for the effects involved in an interaction, an ODDSRATIO statement can be used. An EFFECTSPLOT statement can help display the interaction, as well.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

8. Right-click on the previous Logistic Regression task and select **Add as Code Template** from the drop-down menu. This creates an editable SAS code file to the EGBS project.



9. Open the code file by double-clicking on its icon in the process flow.

10. Scroll down to the SAS code for PROC LOGISTIC.

11. Add the following lines of code to the space just above **RUN;**.

```
ODDSRATIO Gender / at (Class=ALL) cl=pl;
ODDSRATIO Class / at (Gender=ALL) cl=pl;
ODDSRATIO Age / cl=pl;
```

Selected statement:

ODDSRATIO
: produces odds ratios for a variable even when the variable is involved in interactions with other covariates, and for classification variables that use any parameterization. You can also specify variables on which constructed effects are based, in addition to the names of COLLECTION or MULTIMEMBER effects.

Selected options for the ODDSRATIO statement:

AT
: specifies fixed levels of the interacting covariates. If a specified covariate does not interact with the variable, then its AT list is ignored. For continuous interacting covariates, you can specify one or more numbers in the value-list. For classification covariates, you can specify one or more formatted levels of the covariate enclosed in single quotation marks (for example, A='cat' 'dog'), you can specify the keyword REF to select the reference-level, or you can specify the keyword ALL to select all levels of the classification variable. By default, continuous covariates are set to their means, while CLASS covariates are set to ALL.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

CL=PL specifies whether to create Wald or profile-likelihood confidence limits, or both. By default, Wald confidence limits are produced.

The code will look like this:

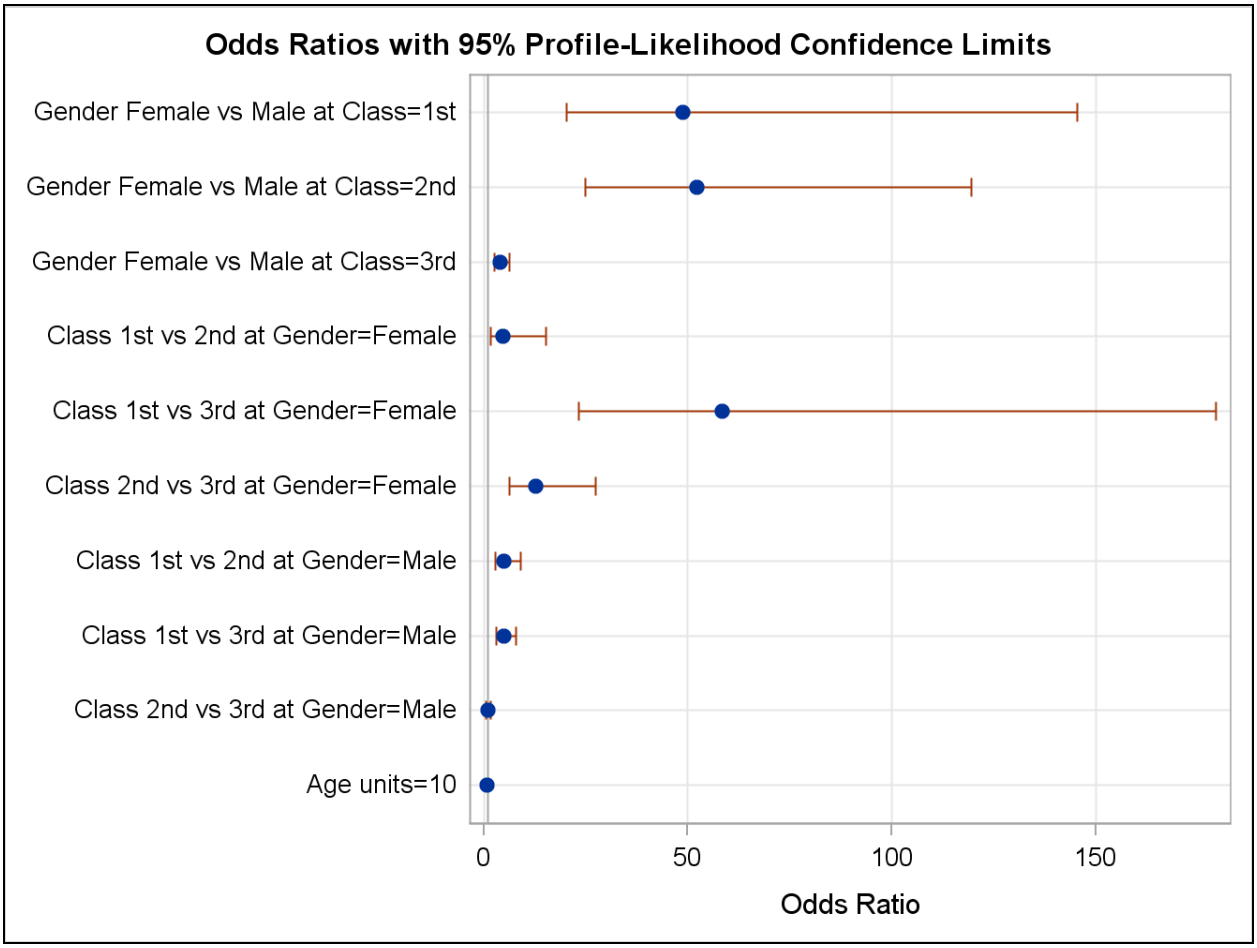
```
PROC LOGISTIC DATA=WORK.SORTTempTableSorted
           PLOTS(ONLY)=ODDSRATIO
           PLOTS(ONLY)=EFFECT
      ;
      CLASS Gender    (PARAM=REF) Class     (PARAM=REF);
      MODEL Survival (Event = 'Survived')=Age Gender Class Age*Gender
Age*Class Gender*Class          /
           SELECTION=BACKWARD
           SLS=0.01
           INCLUDE=0
           LINK=LOGIT
           CLODDS=PL
           ALPHA=0.05
      ;
      UNITS Age  =10;
     ODDSRATIO Gender / at (Class=ALL) cl=pl;
     ODDSRATIO Class / at (Gender=ALL) cl=pl;
     ODDSRATIO Age / cl=pl;
RUN;
QUIT;
```

12. Submit this code by clicking **Run** at the top of the code window.

Partial PROC LOGISTIC Output

Odds Ratio Estimates and Profile-Likelihood Confidence Intervals			
Label	**Estimate**	**95% Confidence Limits**	
Gender Female vs Male at Class=1st	48.735	20.313	145.458
Gender Female vs Male at Class=2nd	52.295	24.869	119.543
Gender Female vs Male at Class=3rd	4.043	2.630	6.253
Class 1st vs 2nd at Gender=Female	4.621	1.590	15.350
Class 1st vs 3rd at Gender=Female	58.349	23.371	179.396
Class 2nd vs 3rd at Gender=Female	12.626	6.330	27.317
Class 1st vs 2nd at Gender=Male	4.959	2.778	9.106
Class 1st vs 3rd at Gender=Male	4.841	2.960	7.978
Class 2nd vs 3rd at Gender=Male	0.976	0.564	1.645
Age units=10	0.681	0.595	0.775

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



From this plot it is clear that the gender effect is different at different classes. (There is a stark difference in first and second classes, but not nearly as much in third class.) The class differences are also evidently different at each gender. For example, the odds ratio comparing first and third classes for females is far greater than the odds ratio comparing those same classes for males.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.